MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

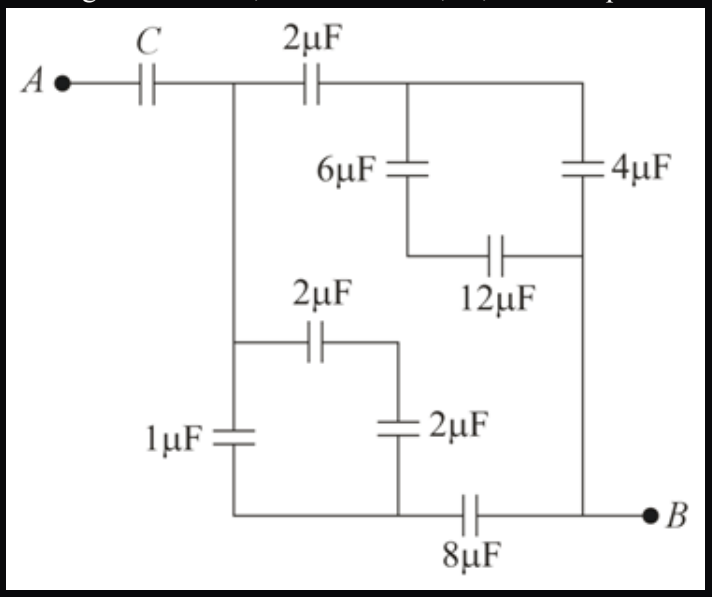
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

In the given network, the value of C, so, that an equivalent capacitance between A and B is 3 μF is:



(A) $36\ \mu F$

(B) $48\ \mu F$

(C) $\frac{31}{5}\ \mu F$

(D) $\frac{1}{5}\ \mu F$

Q.2

The percentage errors in the measurement of mass and speed are 2 % and 3 %, respectively. How much will be the maximum error in the estimation of Kinetic Energy obtained by measuring mass and speed?

(A) $5\ \%$

(B) $1\ \%$

(C) $8\ \%$

(D) $11\ \%$

Q.3

Which of the following pairs of physical quantities does not have same dimensional formula ?

(A) Work and torque

(B) Angular momentum and Planck's constant

(C) Tension and surface tension

(D) Impulse and linear momentum

Q.4

A metal wire of resistance 3Ω is elongated to make a uniform wire of double its previous length. This new wire is now bent and the ends joined to make a circle. If two points on this circle make an angle 60° at the center, the equivalent resistance between these two points will be:

(A) $\frac{5}{3}\Omega$

(B) $\frac{12}{5}\Omega$

(C) $\frac{7}{2}\Omega$

(D) $\frac{5}{2}\Omega$

Q.5

Given below are two statements
Statement I: In hydrogen atom, the frequency of radiation emitted when an electron jumps from lower energy orbit (E_1) to higher energy orbit (E_2), is given as $hf = \mathrm{E}_1 - \mathrm{E}_2$
Statement II: The jumping of electron from higher energy orbit (E_2) to lower energy orbit (E_1) is associated with frequency of radiation given as $f = \frac{(\mathrm{E}_2 - \mathrm{E}_1)}{h}$. This condition is Bohr's frequency condition.
In the light of the above statements, choose the correct answer from the options given below:

(A) Both statement I and statement II are true.

(B) Both statement I and statement II are false.

(C) Statement I is correct but statement II is false.

(D) Statement I is incorrect but statement II is true.

Q.6

The gravitational force between two bodies is directly proportional to $\frac{1}{R}$ not $\left(\frac{1}{R^2}\right)$ where R is the distance between the bodies. Then the orbital speed for this force in a circular orbit is proportional to

(A) $\frac{1}{R^2}$

(B) R^0

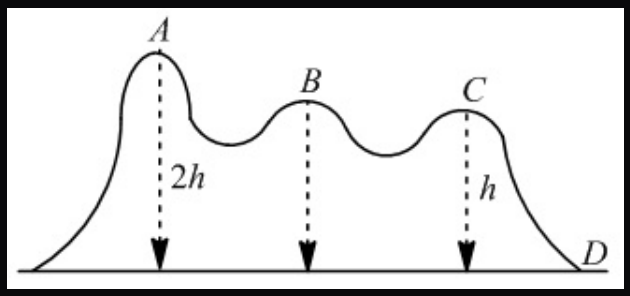
(C) R

(D) $\frac{1}{R}$

Q.7

A small roller coaster starts at point A with a speed u on a curved track as shown in the figure. The friction between the roller

coaster and the track is negligible and it always remains in contact with the track. The speed of roller coaster at point D on the track will be



(A) $\left(u^2+gh\right)^{1/2}$

(B) $\left(u^2+2gh\right)^{1/2}$

(C) $\left(u^2+4gh\right)^{1/2}$

(D) u

Q.8

If the work functions of three photosensitive materials are $1\ \text{eV},\ 2\ \text{eV}$ and $3\ \text{eV}$ respectively, then the ratio of the respective frequencies of light that produce photoelectrons of the maximum kinetic energy of $1\ \text{eV}$ from each of them is

(A) $1:2:3$

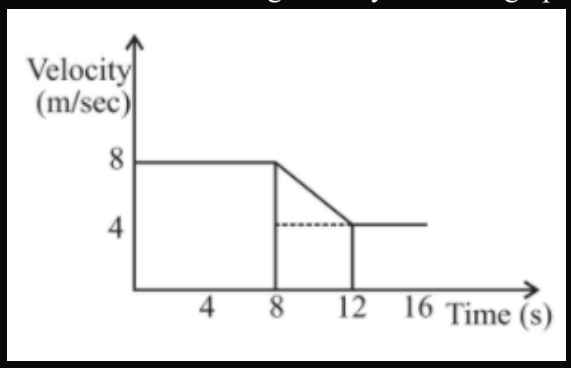
(B) $2:3:4$

(C) $1:1:1$

(D) $3:2:1$

Q.9

Consider the following velocity and time graph.



Which one of the following is the value of average acceleration from 8 seconds to 12 seconds?

(A) $8\ \text{m s}^{-2}$

(B) $12\ \text{m s}^{-2}$

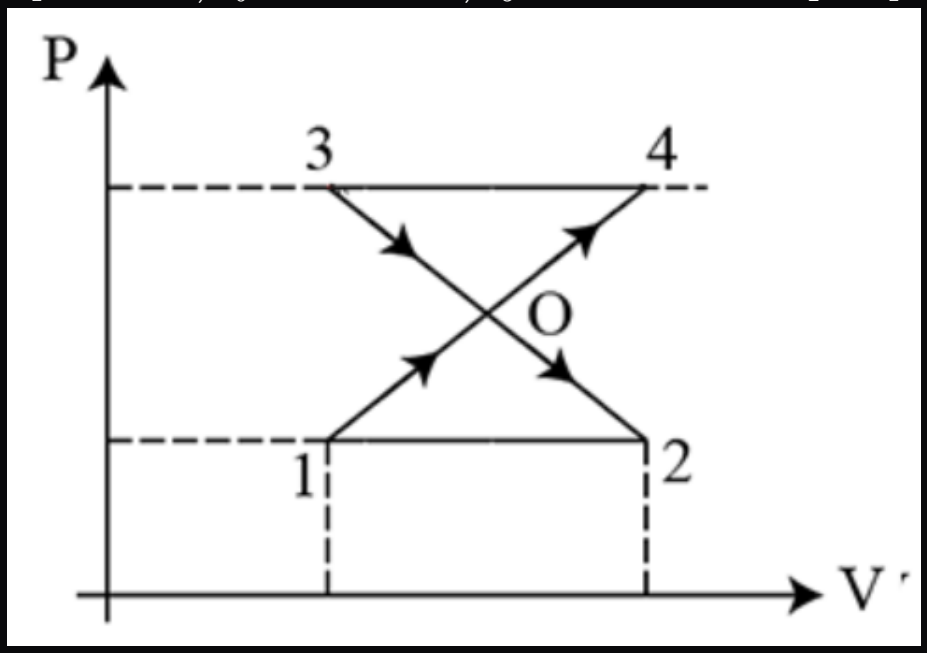
(C) $2\ \text{m s}^{-2}$

(D) $-1\ \text{m s}^{-2}$

Q.10

Determine the work done by an ideal gas undergoing a cyclic process from $1 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$. Given

$P_1 = 10^5$ Pa, $P_0 = 3 \times 10^5$ Pa, $P_3 = 4 \times 10^5$ Pa and $V_2 - V_1 = 10$ L



(A) 740 J

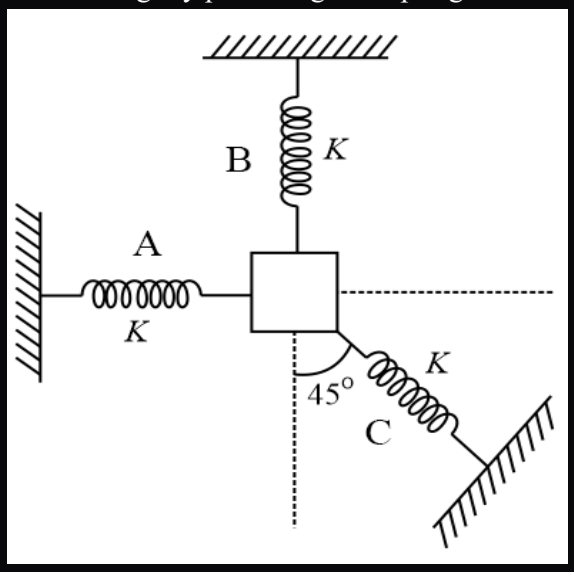
(B) 750 J

(C) 730 J

(D) 745 J

Q.11

A block of mass M is attached to three springs, $A,\ B$ & C having force constants $K,\ K$ and K as shown in the figure. If the block is slightly pushed against spring C. Then find the frequency of oscillations,



(A) $f = 2\pi\sqrt{\frac{M}{K}}$

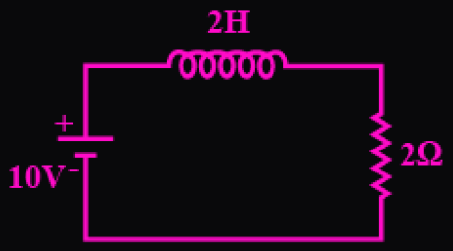
(B) $f = \frac{1}{2\pi}\sqrt{\frac{K}{M}}$

(C) $f = \frac{1}{2\pi}\sqrt{\frac{2K}{M}}$

(D) $f = \frac{1}{2\pi}\sqrt{\frac{3K}{M}}$

Q.12

In the figure magnetic energy stored in the coil is (in steady state)



(A) Zero

(B) Infinite

(C) 25 joules

(D) None of these

Q.13

Four bodies $P,\ Q,\ R$ and S are projected with equal velocities, having angles of projection $15^\circ,\ 30^\circ,\ 45^\circ$ and 60° with the horizontal respectively. The body having the shortest range is

(A) P

(B) Q

(C) R

(D) S

Q.14

One mole of an ideal monoatomic gas at temperature T_0 expands slowly according to the law $\frac{P}{V}=$ constant. If the final temperature is $2T_0$, heat supplied to the gas is

(A) $2RT_0$

(B) RT_0

(C) $\frac{3}{2}RT_0$

(D) $\frac{1}{2}RT_0$

Q.15

For series LCR circuit, the wrong statement is

(A) Applied e.m.f. and potential difference across resistance are in same phase.

(B) Applied e.m.f. and potential difference at inductor coil have phase difference of $\frac{\pi}{2}$.

(C) Potential difference at capacitor and inductor have phase difference of $\frac{\pi}{2}$.

(D) Potential difference across resistance and capacitor have phase difference of $\frac{\pi}{2}$.

Q.16

Statement -1: The sensitivity of a moving coil galvanometer is increased by placing a suitable magnetic materials a core inside the coil.
Statement -2: Soft iron has a high magnetic permeability and cannot be easily magnetized or demagnetized.

(A) Statement -1 is true, Statement-2 is true; Statement -2 is a correct explanation for Statement-1

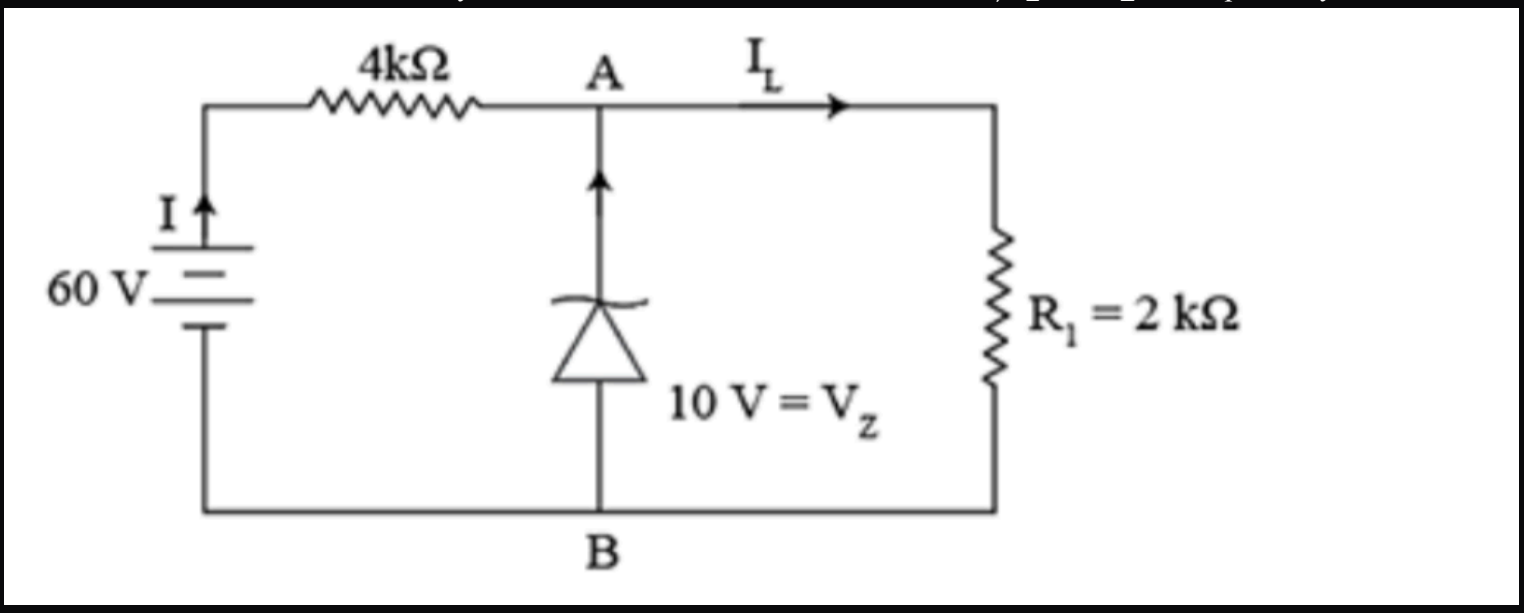
(B) Statement -1 is true, Statement-2 is true; Statement -2 is not explanation for Statement-1

(C) Statement -1 is true, Statement-2 is false

(D) Statement -1 is false, Statement-2 is true

Q.17

A Zener diode is connected to a battery and a load as shown below: The currents I, I_Z and I_L are respectively



(A) $12.5\ mA, 7.5\ mA, 5\ mA$

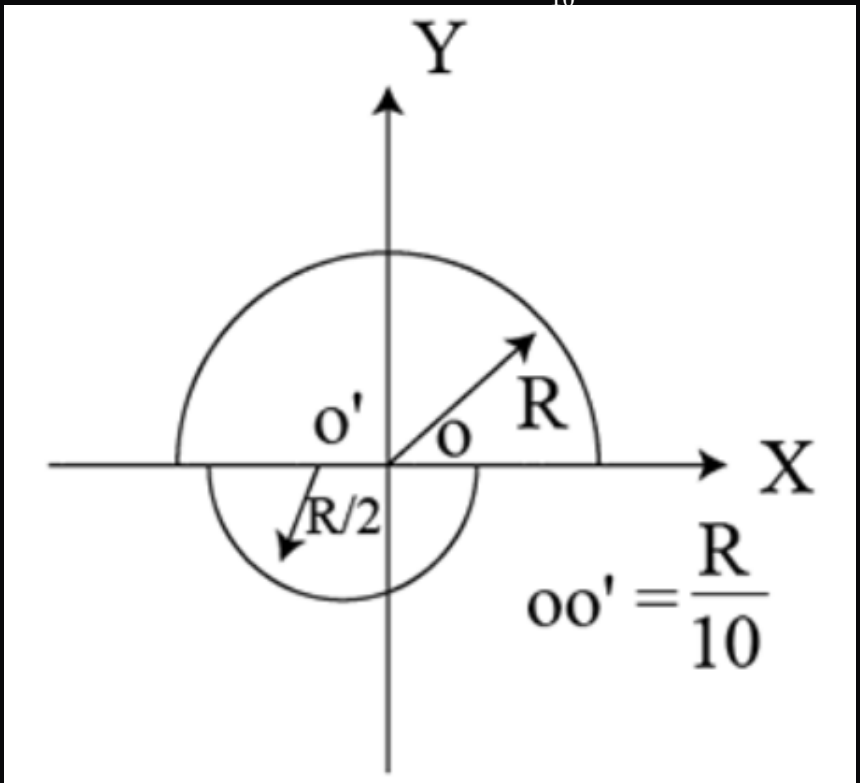
(B) $15\ mA, 7.5\ mA, 7.5\ mA$

(C) $12.5\ mA, 5\ mA, 7.5\ mA$

(D) $15\ mA, 5\ mA, 10\ mA$

Q.18

Two solid hemispheres of radii R and $\frac{R}{2}$ with centers O and O' respectively as shown in figure. The density of bigger hemisphere is ρ and that of smaller hemisphere is 2ρ. Taking center of bigger hemisphere is at origin and the distance between centers of two hemispheres OO' is $\frac{R}{10}$ find coordinates of center of mass of the system.



(A) $\left(-\frac{R}{50}, \frac{21R}{80}\right)$

(B) $\left(-\frac{R}{30}, \frac{21R}{80}\right)$

(C) $\left(-\frac{R}{50}, \frac{7R}{16}\right)$

(D) $\left(-\frac{R}{30}, \frac{7R}{16}\right)$

Q.19

An air bubble of radius 0.1 cm is in a liquid having surface tension 0.06 N/m and density 10^3 kg /m^3. The pressure inside the bubble is 1100 N m^{-2} greater than the atmospheric pressure. At what depth is the bubble below the surface of the liquid? (g = 9.8 m s^{-2})

(A) 0.15 m

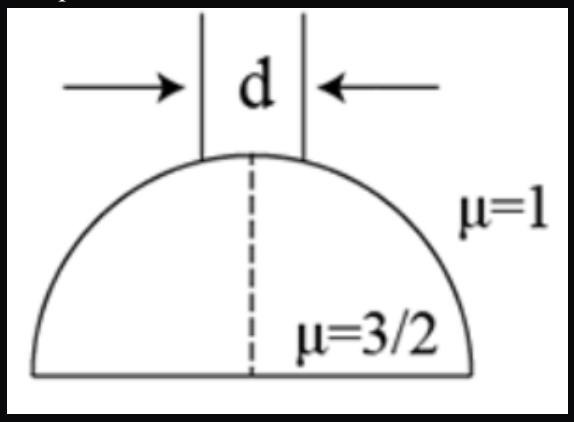
(B) 0.20 m

(C) 0.25 m

(D) 0.1 m

Q.20

A beam of diameter d is incident on a glass hemisphere as shown. If the radius of curvature of the hemisphere is very large in comparison to d, then the diameter of the beam at the base of the hemisphere will be



(A) $\frac{3}{4}d$

(B) d

(C) $\frac{d}{3}$

(D) $\frac{2}{3}d$

Q.21

When two progressive waves $y_1 = 4\sin(2x - 6t)$ and $y_2 = 3\sin\left(2x - 6t - \frac{\pi}{2}\right)$ are superimposed, the amplitude of the resultant wave is

Q.22

The binding energy per nucleon for deuterons ($^2_1$H) and helium ($^4_2$He) are 1.1 MeV and 7.0 MeV respectively. The energy released when two deuterons fuse to form a helium nucleus ($^4_2$He) is : K, find 10 K. Mark answer to nearest integer in Mev

Q.23

The coercivity of a small bar magnet is 4×10^3 A m^{-1} . It is inserted inside a solenoid of 500 turns and length 1 m to demagnetize it. The amount of current to be passed through the solenoid will be _____ A

Q.24

A long solenoid of the cross-sectional radius R has a thin insulated wire ring tightly put on its winding. One half of the ring has the resistance 10 times that of the other half. The magnetic induction produced by the solenoid varies with time as $B = bt$, whose b is a constant. If the magnitude of the electric field strength in the ring is $\frac{p}{q}Rb$, where p and q are the smallest positive integers, then what is the value of $(p+q)$?

Q.25

A particular force F applied on a wire increases its length by 2×10^{-3} m. If we were to increase the wire's length by 4×10^{-3} m, then the applied force is nF. What is the value of n?

Q.26

Which one of the following molecules will form zwitter ion?

(A) $\mathrm{CH_3\,COOH}$

(B) $\mathrm{CH_3\,CH_2\,NH_2}$

(C) $\mathrm{CCl_3\,NO_2}$

(D) $\mathrm{NH_2\,CH_2\,COOH}$

Q.27

We have taken two electrode, with potential values: $\mathrm{E^a_{Mg^{2+}/Mg}} = -2.37\ \mathrm{V}$ and $\mathrm{E^b_{Fe^{3+}/Fe}} = -0.04\ \mathrm{V}$. Which of the following will act as the best reducing agent:

(A) $\mathrm{Mg^{2+}}$

(B) $\mathrm{Fe^{3+}}$

(C) Mg

(D) Fe

Q.28

Which of the following combination of statements is true regarding the interpretation of the atomic orbitals?
(a) An electron in an orbital of high angular momentum stays away from the nucleus than an electron in the orbitals of lower angular momentum.
(b) For a given value of the principal quantum number, the size of the orbit is inversely proportional to the Azimuthal quantum number.
(c) According to wave mechanics, the ground state angular momentum is equal to $\frac{\mathrm{h}}{2\pi}$.
(d) The plot of Ψ Vs r for various azimuthal quantum numbers, show peak shifting towards a higher value.

(A) $\mathrm{(b), (c)}$

(B) $(a), (c)$

(C) $(a), (d)$

(D) $(a), (b)$

Q.29

Which statements of the following is incorrect.

(A) Molecularity of a reaction is always a whole number

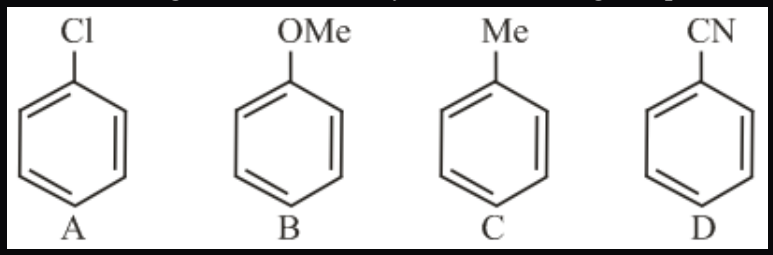
(B) Order and molecularity of a reaction need not be same

(C) Order of a reaction may be zero

(D) Order of a reaction doesn't depends upon the mechanism of the reaction

Q.30

The increasing order of reactivity of the following compounds towards aromatic electrophilic substitution reaction is:



(A) $A < B < C < D$

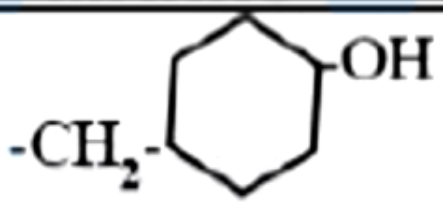
(B) $B < C < A < D$

(C) $D < B < A < C$

(D) $D < A < C < B$

Q.31

Match the amino acid with its correct R group.

	Amino acid		R- group
P	Cysteine	I	$-CH_2OH$
Q	Serine	II	$-CH_2-CH_2-CH_2-COOH$
R	Glutamic acid	III	
S	Tyrosine	IV	$-CH_2-SH$

(A) $P-I,\ Q-IV,\ R-II,\ S-III$

(B) $P-IV,\ Q-I,\ R-II,\ S-III$

(C) $P-III,\ Q-IV,\ R-II,\ S-I$

(D) $\mathrm{P-II,\ Q-I,\ R-IV,\ S-III}$

Q.32

In the titration of $\mathrm{I_2\ (aq)}$ by $\mathrm{S_2O_3^{2-}\ (aq)}$ using the starch indicator, the end point is indicated by:

(A) Colourless to blue

(B) Blue to colourless

(C) Pink to colourless

(D) Blue to pink

Q.33

$\mathrm{O_2}$ molecule is paramagnetic due to presence of :

(A) Two unpaired electrons in antibonding MO

(B) One unpaired electron in antibonding MO

(C) Two unpaired electrons in bonding MO

(D) One unpaired electron in bonding MO

Q.34

Mark out the enthalphy of formation of carbon monoxide (CO)
Given, $\mathrm{C}(s)+\mathrm{O_2(g)} \rightarrow \mathrm{CO_2(g)},\ \ \Delta\,\mathrm{H}=-393.3\ \mathrm{kJ/mol}$ & $\mathrm{CO_{(g)}+\frac{1}{2}O_{2(g)} \rightarrow CO_{2(g)}; \Delta H=-282.2\ KJm^{-1}}$

(A) $-111.1\ \mathrm{KJ\ mol^{-1}}$

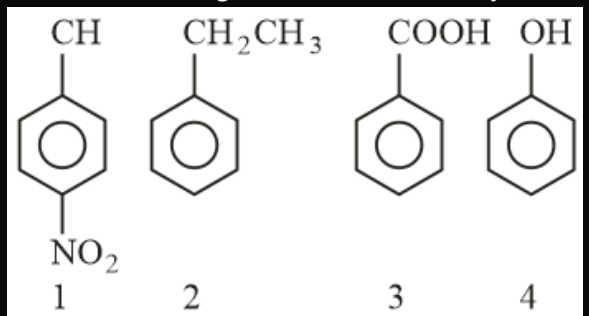
(B) $676.1\ \mathrm{KJ\ mol^{-1}}$

(C) $-282.8\ \mathrm{KJ\ mol^{-1}}$

(D) $300.0\ \mathrm{KJ\ mol^{-1}}$

Q.35

Which will undergo a Friedel-Craft's alkylation reaction?



(A) $1,\ 2$ and $4.$

(B) 1 and $3.$

(C) 2 and $4.$

(D) 1 and $2.$

Q.36

Statement - 1: Only temperature can change the vapour pressure of a pure liquid.
Statement - 2: Equilibrium constant does not change unless temperature is changed.

(A) Statement-1, is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement-1.

(B) Statement - 1 is True, Statement is True; Statement - 2, is NOT a correct explanation for Statement - 1.

(C) Statement - 1 is True, Statement - 2, is False.

(D) Statement - 1 is False, Statement - 2 is True.

Q.37

Arrange the following anilines in decreasing order of basicity
1. $C_6H_5NH_2$
2. o $-$ $CH_3C_6H_4NH_2$
3. m $-$ $CH_3C_6H_4NH_2$
4. p $-$ $CH_3C_6H_4NH_2$

(A) 4 > 1 > 2 > 3

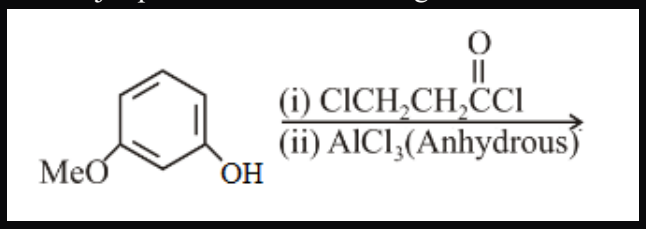
(B) 4 > 3 > 1 > 2

(C) 1 > 2 > 3 > 4

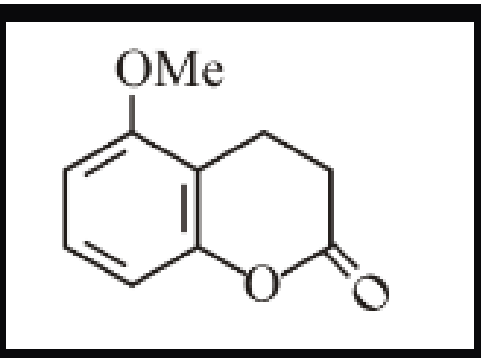
(D) 4 > 3 > 2 > 1

Q.38

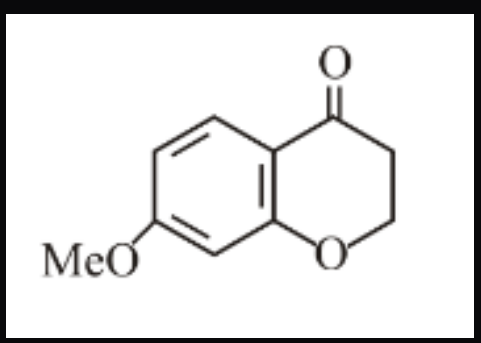
The major product of the following reaction is:



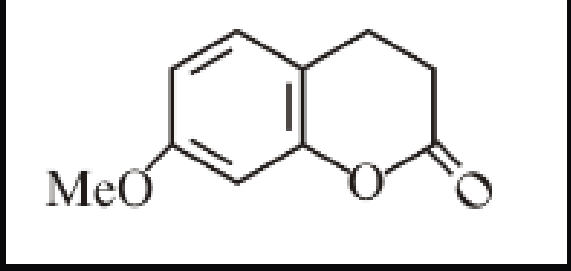
(A)



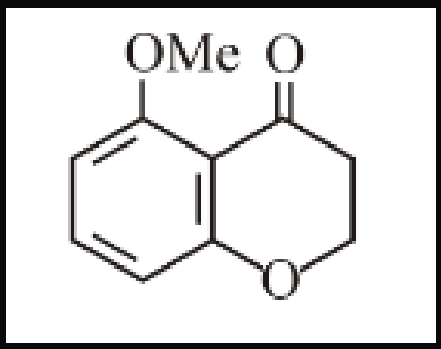
(B)



(C)



(D)



Q.39

Which of the following statement about cerium is not true?

(A) The common oxidation states of cerium are $+3$ and $+4$.

(B) The $+3$ oxidation state of cerium is more stable than $+4$ oxidation state.

(C) Cerium (III) acts as an oxidising agent.

(D) The $+4$ oxidation state of cerium is known in solution.

Q.40

Addition of excess aqueous ammonia to a pink coloured aqueous solution of $\mathrm{MCl_2.6H_2O(X)}$ and $\mathrm{NH_4Cl}$ gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as $1:3$ electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 B.M., whereas it is zero for complex Y. Among the following options, which statement is incorrect?

(A) The hybridization of the central metal ion in Y is $\mathrm{d^2sp^3}$

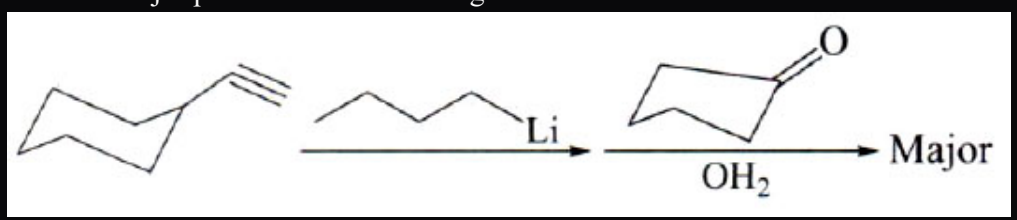
(B) Addition of silver nitrate to Y gives only two equivalents of silver chloride

(C) When X and Z are in equilibrium at $0^\circ\mathrm{C}$, the colour of the solution is pink

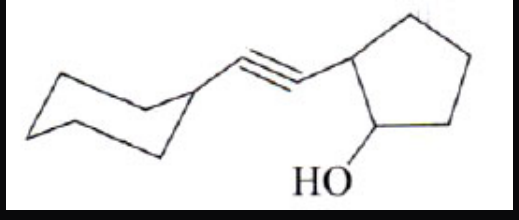
(D) Z is a tetrahedral complex

Q.41

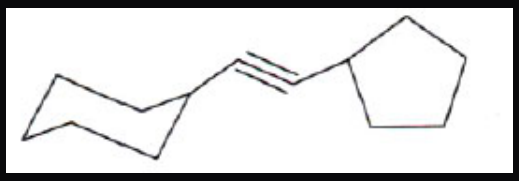
Give the major product of the following reaction



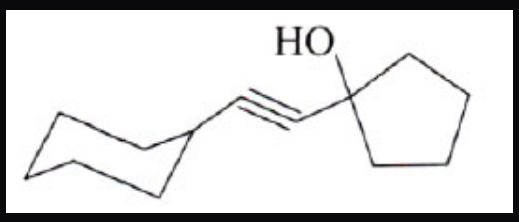
(A)



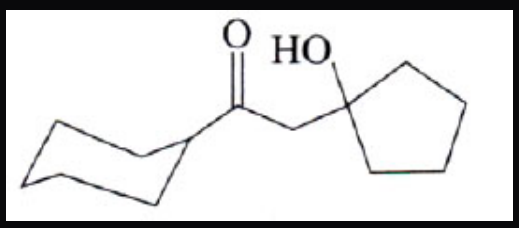
(B)



(C)



(D)



Q.42

4ml of HCl solution of $\text{pH} = 2$ is mixed with 6ml of NaOH solution of $\text{pH} = 12$. What would be the final pH of solution? $(\log 2 = 0.3)$

(A) 10.3

(B) 11.3

(C) 11

(D) 4.3

Q.43

A: The position of hydrogen atom in the modern periodic table is group 1 of the 1st period only
R: Hydrogen atom has one proton with one neutron.

(A) If both Assertion & Reason are true and the reason is the correct explanation of the assertion, then mark (1).

(B) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion, then mark (2).

(C) If Assertion is true statement but Reason is false, then mark (3).

(D) If both Assertion and Reason are false statements, then mark (4).

Q.44

A: Hybridization of carbon is sp^2 in all its crystalline allotropes.
R: There are alternate double-single bonds in each allotrope of carbon.

(A) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.

(B) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(C) Assertion is true statement but Reason is false.

(D) Both Assertion and Reason are false statements.

Q.45

Select the correct statement.

(i) Delocalisation of σ-electron is hyperconjugation.

(ii) Delocalisation of π-electron is resonance.

(iii) Partial displacement of σ-electron is inductive effect.

(A) i & iii

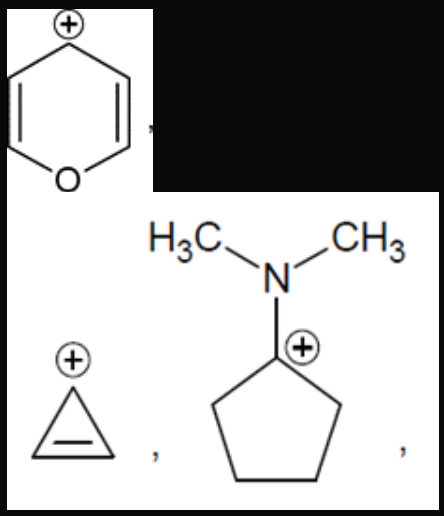
(B) ii & iii

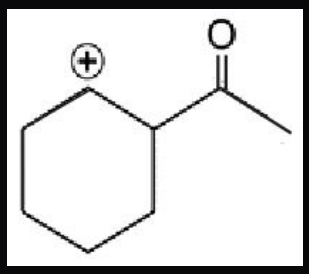
(C) i & ii

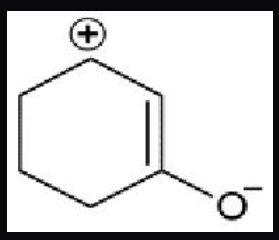
(D) i, ii & iii

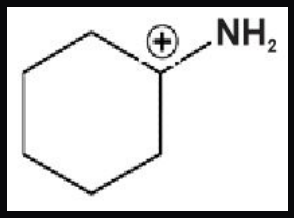
Q.46

How many of the following intermediates have at least one contributing structure in which all atoms have their octet complete?





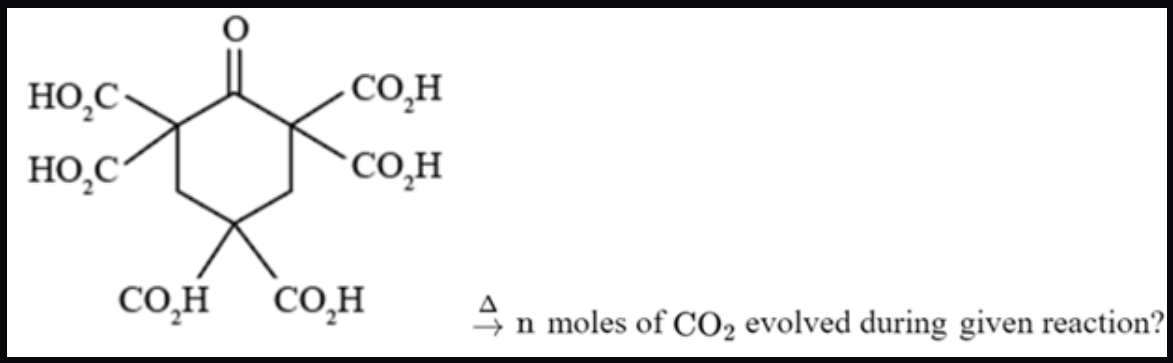




Q.47

Among the complex ions,
$[Co(NH_2-CH_2-CH_2-NH_2)_2\ Cl_2]^+$, $[CrCl_2(C_2O_4)_2]^{3-}$, $[Fe(H_2O)_4(OH)_2]^+$, $[Fe(NH_3)_2](CN)_4]$
and $[Co(NH_3)_4(H_2O)Cl]^{2+}$, the number of complex ion(s) that show(s) cistrans isomerism is

Q.48



$\xrightarrow{\Delta}$ n moles of CO_2 evolved during given reaction?

What is the value of ' n ' here?

Q.49

Standrad entropies of X_2, Y_2 and XY_3 are 60, 40 and 50 $JK^{-1}mol^{-1}$ respectively. For the reaction
$\frac{1}{2}X_2+\frac{3}{2}Y_2 \leftrightarrow XY_3, \Delta H=-30$ kJ to be at equilibrium, the temperature should be ____ K

Q.50

A carbonyl compound of formula $C_9H_{10}O(A)$, which is a benzene derivative gives orange precipitate with 2,4 -D.N.P. and also gives yellow precipitate with I_2 in presence of aqueous NaOH. The total no. of isomers possible for 'A' are_____

Q.51

The general solution of the differential equation $\frac{dy}{dx}=2y\tan x+\tan^2 x,\ \forall x\in\left(0,\frac{\pi}{2}\right)$ is $yf(x)=\frac{x}{2}-\frac{\sin(2x)}{4}+C$, (where, C is an arbitrary constant). If $f\left(\frac{\pi}{4}\right)=\frac{1}{2}$, then the value of $f\left(\frac{\pi}{3}\right)$ is equal to

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) 2

(D) 4

Q.52

Let the vectors $\vec{a},\vec{b}$ and $\vec{c}$ are perpendicular to $\vec{b}+\vec{c},\vec{c}+\vec{a}$ and $\vec{a}+\vec{b}$ respectively. If $\left|\vec{a}+\vec{b}\right|=6$, $\left|\vec{b}+\vec{c}\right|=8$ and $\left|\vec{c}+\vec{a}\right|=10$, then the value of $\left|\vec{a}+\vec{b}+\vec{c}\right|$ is equal to

(A) $5\sqrt{5}$

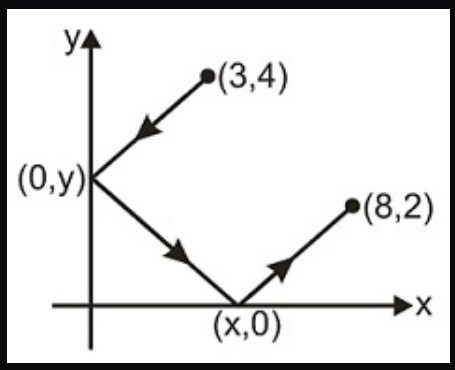
(B) 50

(C) $10\sqrt{2}$

(D) 10

Q.53

Suppose that a ray of light leaves the point $(3, 4)$, reflects off the y - axis towards the x - axis, reflects off the x - axis, and finally arrives at the point $(8, 2)$. The value of x, is :



(A) $x = 4\frac{1}{2}$

(B) $x = 4\frac{1}{3}$

(C) $x = 4\frac{2}{3}$

(D) $x = 5\frac{1}{3}$

Q.54

If $(1+x)^n = \sum_{r=0}^{n} a_r x^r$ and $b_r = 1 + \frac{a_r}{a_{r-1}}$and $\prod_{r=1}^{n} b_r = \frac{(101)^{100}}{100!}$, then n equals to

(A) 99

(B) 100

(C) 101

(D) 102

Q.55

The shortest distance between the two lines $L_1 : x = k_1;\ y = k_2$ & $L_2 : x = k_3;\ y = k_4$ is equal to

(A) $\left|\sqrt{k_1^2 + k_2^2} - \sqrt{k_3^2 + k_4^2}\right|$

(B) $\sqrt{k_1 k_3 + k_2 k_4}$

(C) $\sqrt{(k_1 + k_3)^2 + (k_2 + k_4)^2}$

(D) $\sqrt{(k_1 - k_3)^2 + (k_2 - k_4)^2}$

Q.56

If a, b, c be the pth, q^{th} and r^{th} terms respectively of an AP and GP both, then the product of the roots of the equation $a^b b^c c^a\, x^2 - abcx + a^c b^a c^b = 0$ is equal to-

(A) -1

(B) 1

(C) 2

(D) $(\mathrm{b}-\ \mathrm{c})\,(\mathrm{c}-\ \mathrm{a})\,(\mathrm{a}-\ \mathrm{b})$

Q.57

If $y(x)$ is a solution of $\frac{dy}{dx}-\frac{xy}{1+x}=\frac{1}{1+x}$ and $y(0)=-1$, then the value of $y(2)$ is

(A) $-\frac{1}{2}$

(B) $-\frac{1}{3}$

(C) $-\frac{1}{4}$

(D) $-\frac{1}{5}$

Q.58

Let $f: R \to R$ such that $f(x+2y)=f(x)+f(2y)+4xy, x, y \in R$ and $f'(0)=0$
If $I_1=\int_0^1 f(x)dx, I_2=\int_{-1}^0 f(x)dx, I_3=\int_{\sqrt{1/2}}^2 f(x)dx$, then

(A) $I_1=I_2>I_3$

(B) $I_1>I_2>I_3$

(C) $I_1=I_2<I_3$

(D) $I_1<I_2<I_3$

Q.59

Let f: R $\to$ R and $f_n(x)=f(f_{n-l}(x))\ \forall n \geq 2,\ n \in N$. Then the roots of the equation $f_3(x)f_2(x)f(x)-25f_2(x)f(x)$ $+175f(x)=375$ which also satisfy the equation $f(x)=x$ will be

(A) 5 only

(B) 15 only

(C) 10 only

(D) $5,\ 15$

Q.60

The probability distribution of a random variable (X) is $P(X)=\begin{cases}\frac{X}{12} & : X=1,2,3,4,5,6\\ 0 & : \text{otherwise}\end{cases}$. Then, the conditional probability $P\left(\frac{\frac{3}{2}<X<\frac{7}{2}}{X>2}\right)$ is

(A) $\frac{5}{6}$

(B) $\frac{5}{18}$

(C) $\frac{1}{6}$

(D) $\frac{7}{12}$

Q.61

Let $f(b)$ be the minimum value of the expression $y = x^2 - 2x + \left(b^3 - 3b^2 + 4\right)\ \forall x \in R$. Then, the maximum value of $f(b)$ as b varies from 0 to 4 is

(A) 20

(B) 19

(C) 63

(D) 64

Q.62

The value of expression $\frac{\sin^4\alpha+\cos^4\alpha+2\sin^2\alpha\cos^2\alpha}{\sin^6\alpha+\cos^6\alpha+3\sin^2\alpha\cos^2\alpha} + 1$ is equal to:

(A) 1

(B) 3

(C) 2

(D) 0

Q.63

If $a_1,\ a_2\ ,\ldots,\ a_n$ are real numbers with $a_n \neq 0$ and $\cos\alpha + i\sin\alpha$ is a root of
$z^n + a_1 z^{n-1} + a_2 z^{n-2} + \ldots + a_{n-1} z + a_n = 0$, then the sum $a_1\cos\alpha + a_2\cos 2\alpha + a_3\cos 3\alpha + \ldots + a_n\cos n\alpha$ is

(A) 0

(B) 1

(C) -1

(D) $\frac{1}{2}$

Q.64

The geometric mean of 6 observations was calculated as 13. It was later observed that one of the observations was recorded as 28 instead of 36. The correct geometric mean is

(A) $\left(\frac{9}{7}\right)^{\frac{1}{6}}$

(B) $3\left(\frac{9}{7}\right)^{\frac{1}{6}}$

(C) $13\left(\frac{9}{7}\right)^{\frac{1}{6}}$

(D) $13\left(\frac{7}{9}\right)^{\frac{1}{6}}$

Q.65

The number of ways in which three numbers in arithmetic progression can be selected from $\{1, 2, 3, \ldots\ldots, 50\}$ is

(A) 276

(B) 600

(C) 840

(D) 640

Q.66

If both the mean and the standard deviation of 50 observations $x_1,\ x_2,\ \ldots,\ x_{50}$ are equal to 16, then the mean of $(x_1-4)^2,\ (x_2-4)^2,\ \ldots,(x_{50}-4)^2$ is

(A) 525

(B) 480

(C) 400

(D) 380

Q.67

If A is a square matrix of order $n \times n$ such that $A^2 = A$ and I is a unit matrix of order $n \times n$, then $(I+A)^n$ is equal to

(A) $I + 2^n A$

(B) $I + (2^n - 1)A$

(C) $I - (2^n - 1)A$

(D) None of these

Q.68

The point $\left(\frac{1}{e}, \frac{1}{e'}\right)$ lies on, if e and e' are the eccentricities of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$

(A) $x^2 + y^2 = 1$

(B) $x^2 + y^2 = 2$

(C) $x^2 + y^2 = 3$

(D) $x^2 + y^2 = 4$

Q.69

The area of the region $\{(x,\ y) : x^2 + y^2 \le 1 \le x + y\}$ is

(A) $\frac{\pi^2}{5}$ sq unit

(B) $\frac{\pi^2}{2}$ sq unit

(C) $\left(\frac{\pi}{4} - \frac{1}{2}\right)$ sq unit

(D) $\frac{\pi}{4}$ sq unit

Q.70

The value of the integral $\int \frac{x^2 - 4x\sqrt{x} + 6x - 4\sqrt{x} + 1}{x - 2\sqrt{x} + 1}\,dx$ is equal to

(A) $\frac{x^{\frac{3}{2}}}{2} + x + c$

(B) $\frac{x^2}{2} - \frac{4}{3}x^{\frac{3}{2}} + x + c$

(C) $x^{\frac{3}{2}} + \frac{x}{2} + c$

(D) $\frac{2}{3}x^{\frac{3}{2}} + c$

Q.71

The highest power of 18 contained in ${}^{100}C_{50}$ is

Q.72

If the triangle ABC whose vertices are $A(-1, 1, 1)$, $B(1, -1, 1)$ and $C(1, 1, -1)$ projected on XY-plane, then the area of the projected triangle is

Q.73

$\lim\limits_{h\to\infty} h^{-h^2}\left[(h+1)\left(h+\frac{1}{2}\right)\left(h+\frac{1}{2^2}\right)\ldots\ldots\ldots\ldots\ldots\ldots\left(h+\frac{1}{2^{h-1}}\right)\right]^h = e^s$
then S =.....

Q.74

If the line $\mathrm{y} - 2 = 0$ is the directrix of the parabola $\mathrm{x}^2 - \mathrm{ky} + 32 = 0, \mathrm{k} \neq 0$ and the parabola intersects the circle $\mathrm{x}^2 + \mathrm{y}^2 = 8$ at two real distinct points, then the absolute value of k is

Q.75

If the number of integral values of x satisfying the inequality $\frac{(e^x-1)(\sin x-2)(x^2-5x+4)}{x^2(-x^2+x-2)(2x+3)} \leq 0$ is λ, then λ^2 is equal to

Answers & Solutions

Q.1 Answer:

48 μF

Solution:

First of all see in lower loop, both $2\ \mu\mathrm{F}$ capacitor are in series.

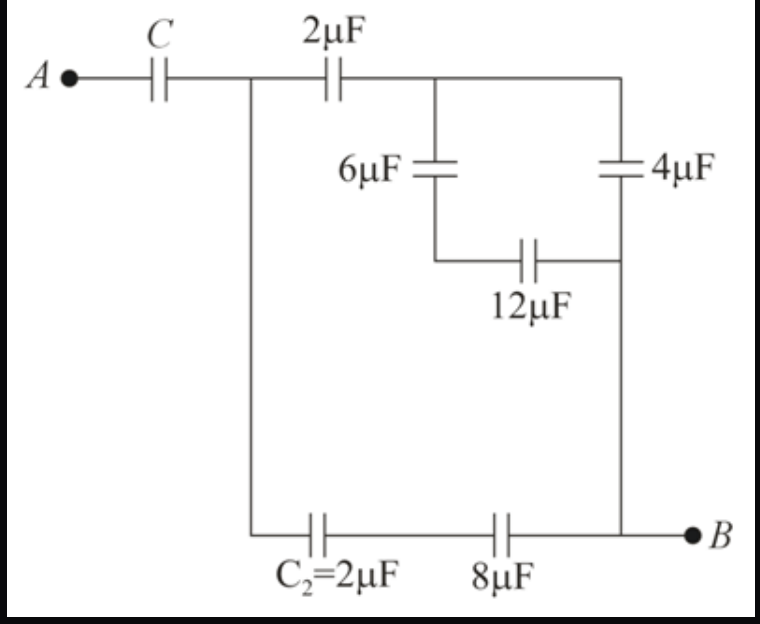
So, their equivalent capacitor of capacitance can be written as,

$\frac{1}{C_1} = \frac{1}{2} + \frac{1}{2} = 1\ \mu\mathrm{F} \Rightarrow C_1 = 1\ \mu\mathrm{F}$

Now, this C_1 is in parallel with $1\ \mu\mathrm{F}$ capacitor

So, equivalent capacitance can now be written as,

$C_2 = 1 + 1\ = 2\ \mu\mathrm{F}$



Now, C_2 is in series with $8\ \mu F$, so, equivalent capacitance can be written as,

$\frac{1}{C_3} = \frac{1}{2} + \frac{1}{8} = \frac{5}{8} C_3 = 1.6\ \mu F$

Moving to the upper loop, $6\ \mu F$ and $12\ \mu F$ are in series. So, equivalent capacitance can be written as,

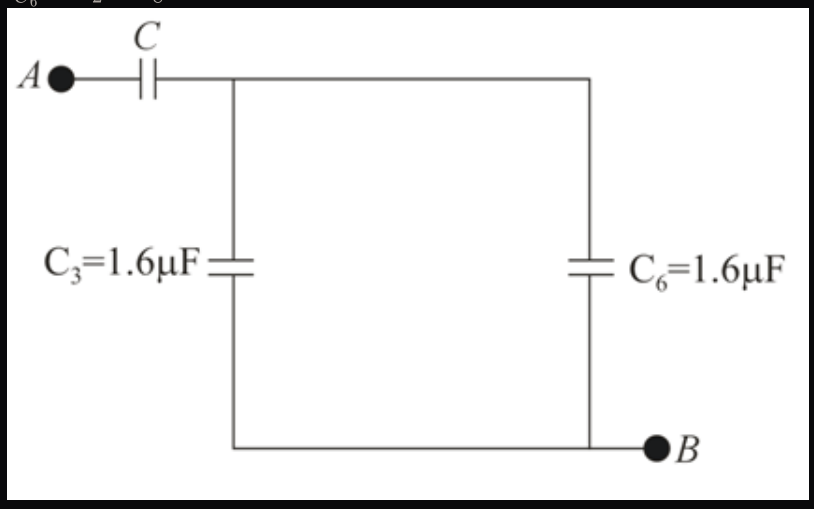
$\frac{1}{C_4} = \frac{1}{6} + \frac{1}{12} C_4 = 4\ \mu F$

Now, this C_4 is in parallel with $4\ \mu F$.So, equivalent capacitance can be written as,

$C_5 = 4 + 4 = 8\ \mu F$

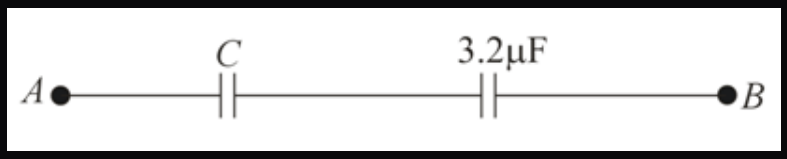
Now, this C_5 and $2\ \mu F$ are in series. So, equivalent cpacitance can be written,

$\frac{1}{C_6} = \frac{1}{2} + \frac{1}{8} \Rightarrow C_6 = 1.6\ \mu F$



Now, as shown in the diagram, C_3 and C_6 are in parallel. So, their equivalent capacitance can be given by,

$C\prime = 1.6 + 1.6 = 3.2\ \mu F$



Finally, this $C\prime$ and $3\ \mu F$ are in series with an equivalent capacitance of $3\ \mu F$ (as given in question):

So, we can write this as,

$\frac{1}{3} = \frac{1}{3.2} + \frac{1}{C}$

Solving this we get,

$C = 48\ \mu F$.

Q.2 Answer:

$8\ \%$

Solution:

The kinetic energy of a particle will be,

$KE = \frac{1}{2}mv^2$,

where m is mass and v is the speed of the particle.

The error in kinetic energy of the body will be,

$\frac{\Delta KE}{KE} \times 100 = \frac{\Delta m}{m} \times 100 + 2\frac{\Delta v}{v} \times 100$

$\frac{\Delta KE}{KE} \times 100 = 2 + 2 \times 3 = 8\ \%$

will be the error in the kinetic energy of particles.

Q.3 Answer:

Tension and surface tension

Solution:

The pairs of physical quantities does not have same dimensional formula tension and surface tension

$[\text{Tension}] = \left[\text{ML}^{\text{-1}}\right]$

$[\text{Surface Tension}] = \left[\text{ML}^{0}\text{T}^{\text{-2}}\right]$

Q.4 Answer:

$\frac{5}{3}\Omega$

Solution:

We know, $R = \frac{8l}{A} = \frac{8l^2}{Al} = \frac{8l^2}{\text{Volume}}$ $R \propto l^2$ **if the volume is constant.**

$\therefore$ **New resistance** $= 4R$

$R_2 = \frac{4R}{360} \times 300$

$$R_1 = \frac{4R}{360} \times 60$$

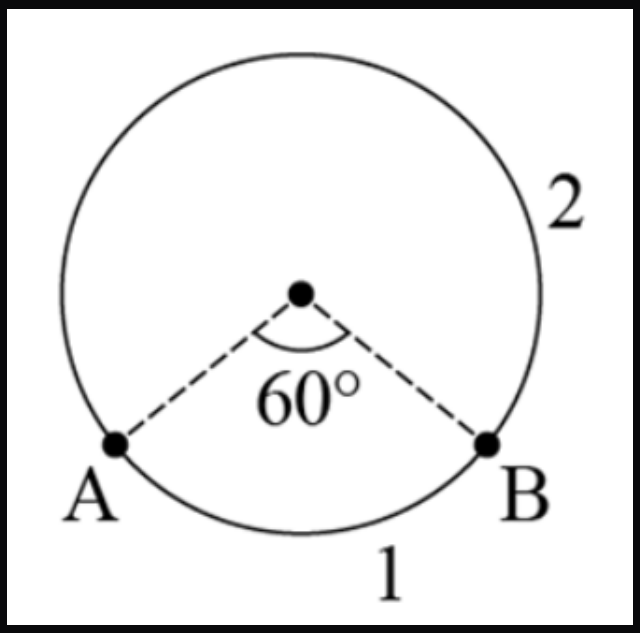
$$R_1 = \frac{4R}{360} \times 300 = \frac{10R}{3}$$

$$R_2 = \frac{4R}{360} \times 60 = \frac{2R}{3}$$

$\therefore$ Equivalent resistance $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$

$$= \frac{\frac{10R}{3} \times \frac{2R}{3}}{\frac{10R}{3} + \frac{2R}{3}} = \frac{5R}{9}$$

$$= \frac{5}{3}\Omega$$



Q.5 Answer:

Statement I is incorrect but statement II is true.

Solution:

When electron jump from lower to higher energy level, energy absorbed so statement-I incorrect. When electron jump from higher to lower energy level, energy of emitted photon

$E = E_2 - E_1$

$hf = E_2 - E_1 \Rightarrow f = E_2 - E_1/h$

so statement-II is correct

Q.6 Answer:

R^0

Solution:

We know that the centripetal force acting on a body in any orbit is provided by gravitational force.

Now according to the question gravitational force between two masses m_1 and m_2 is, $F_G = \frac{Gm_1m_2}{R}$, where G is the gravitational constant, R is separation between them.

Now, let m_1 and m_2 be the mass of two bodies, and m_2 is orbiting around m_1 and $m_2 \gg m_1$.

$\therefore F_G = \frac{Gm_1m_2}{R} = \frac{m_2V^2}{R}$

where V is the orbital velocity.

$\Rightarrow\ V^2 = Gm_1$

$\therefore\ V$ **is proportional to** R^0**.**

Q.7 Answer:

$\left(\mathrm{u}^2 + 4\mathrm{gh}\right)^{1/2}$

Solution:

By the conservation of mechanical energy,

$U + K = cons\tan t \Rightarrow mg(2h) + \frac{1}{2}mu^2 = 0 + \frac{1}{2}mv^2$

$\Rightarrow v = \left(\mathrm{u}^2 + 4\mathrm{gh}\right)^{1/2}$

Q.8 Answer:

$2\ :3\ :4$

Solution:

From Einstein photo electric equation

$hv = KE + \phi$

Where, $\phi =$ **work function** $= hv_0$

$KE = 1\ \mathrm{eV}$

$hv_1 = 1 + 1 = 2\ \mathrm{eV}$

$hv_2 = 1 + 2 = 3\ \mathrm{eV}$

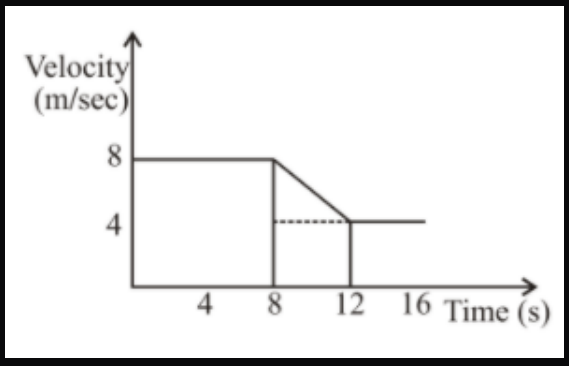
$hv_3 = 1 + 3 = 4\ \mathrm{eV}$

$\Rightarrow\ \ v_1\ : v_2\ : v_3 = 2\ :3\ :4$

Q.9 Answer:

$-1\ \mathrm{m\ s^{-2}}$

Solution:



According to graph, we have

velocity at $A = 8\ \mathrm{m\ s^{-1}}$**,**

velocity at $B = 4\ \mathrm{m\ s^{-1}}$**,**

time-interval $= (12 - 8) = 4$ **seconds.**

The average acceleration from 8 **seconds to** 12 **seconds, is given as**

Average acceleration, $a_{\mathrm{avg}} = \frac{\text{velocity at } B - \text{velocity at } A}{\text{time interval}}$

$a_{\mathrm{avg}} = \frac{4-8}{4} = \frac{-4}{4} = -1$

Hence, the average acceleration is $-1\ \mathrm{m\ s^{-2}}$**.**

Q.10 Answer:

750 J

Solution:

From figure

$$\frac{\mathrm{V_4 - V_3}}{\mathrm{V_2 - V_1}} = \frac{\mathrm{P_3 - P_0}}{\mathrm{P_0 - P_1}} \Rightarrow \frac{\mathrm{V_4 - V_3}}{10} = \frac{4 \times 10^5 - 3 \times 10^5}{3 \times 10^5 - 10^5}$$

$$V_4 - V_3 = 5\ \mathrm{L}$$

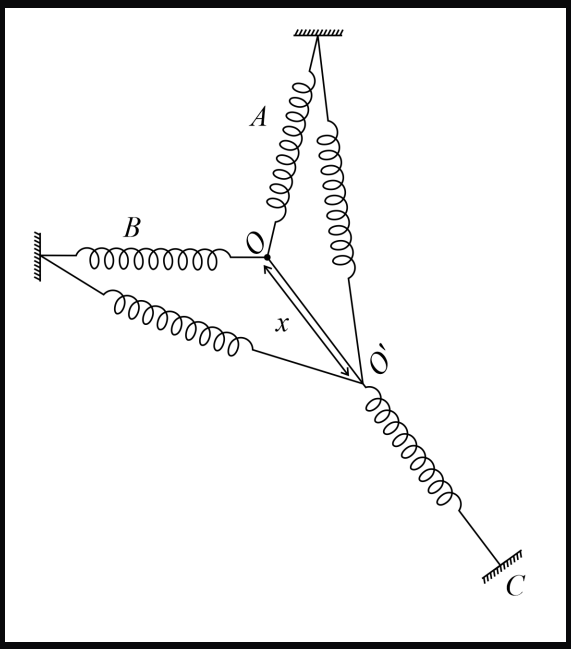
Now, work done

$$\mathrm{W} = \left(\tfrac{1}{2} \times 1 \times 2 \times 10^5 - \tfrac{1}{2} \times 5 \times 1 \times 10^5\right) \times 10^{-3} = 750\ \mathrm{J}$$

Q.11 Answer:

$f = \frac{1}{2\pi}\sqrt{\frac{2\mathrm{K}}{M}}$

Solution:



Suppose the common point is taken x distance along point C.

From the diagram, we can see that extension in the spring A and B,

$x_1 = x\cos 45^\circ = \frac{x}{\sqrt{2}}$.

Extension in spring C,

$x_2 = x$

Force developed in spring A and B,

$F_1 = kx_1 = \frac{kx}{\sqrt{2}}$.

Force developed in every spring C,

$F_2 = kx$

Sum of components of all these forces in the direction of OC will provide restoring force,

$F_{\text{net}=} = -2 \times \left(\frac{kx}{\sqrt{2}}\cos 45^\circ\right) + kx = 2 \times \frac{kx}{2} + kx$

$F_{\text{net}} = -2kx$.

Using newton's second law-

$F_{\text{net}} = ma = -2kx$

$ma = -2kx$

$a = -\left(\frac{2\mathrm{k}}{m}\right)x$

For SHM,

$a = -\omega^2 x = -\left(\frac{2\pi}{T}\right)^2 x$

$\therefore \quad \left(\frac{2\pi}{\mathrm{T}}\right)^2 = \frac{2\mathrm{k}}{m}$

$\Rightarrow T=2\pi\sqrt{\frac{m}{2k}}$

Q.12 Answer:

25 joules

Solution:

$\mathrm{i}=\frac{\mathrm{V}}{\mathrm{R}}=\frac{10}{2}=5\ \mathrm{A}$
$\mathrm{U}=\frac{1}{2}\mathrm{Li}^2=\frac{1}{2}\times 2\times 25=25\ \mathrm{J}$

Q.13 Answer:

P

Solution:

For projectile motion, horizontal range is given by,
$R=\frac{u^2\sin 2\theta}{g}$
Substituting $\theta=45^\circ$ **gives the maximum range.**
$R_R=\frac{u^2\times\sin(2\times 45^\circ)}{g}=\frac{u^2}{g}$
The ranges of projectile for different angles of projection are calculated as follows,
$R_P=\frac{u^2\times\sin(2\times 15^\circ)}{g}=\frac{u^2}{2g}=0.5\frac{u^2}{g}$,
$R_Q=\frac{u^2\times\sin(2\times 30^\circ)}{g}=\frac{\sqrt{3}u^2}{2g}=0.866\frac{u^2}{g}$
and
$R_S=\frac{u^2\times\sin(2\times 60^\circ)}{g}=\frac{\sqrt{3}u^2}{2g}$
Hence, range increases when θ **increases from** 0° **to** 45° **but, it decreases beyond** 45°**. When the angle of projection is very far from** 45°**, then, range will be minimum which is** 15° **for body** P**.**

Q.14 Answer:

$2RT_0$

Solution:

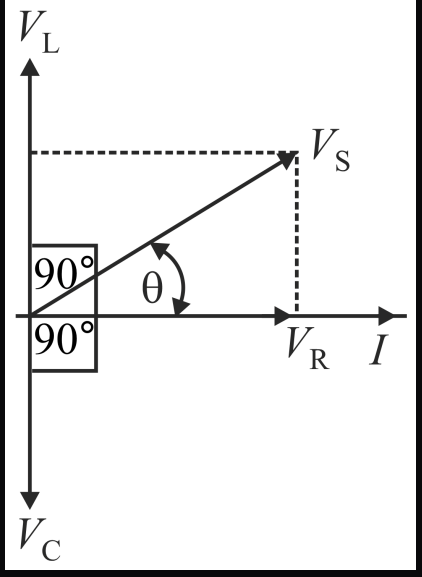
In a process $PV^x=$ **constant, molar heat capacity is given by** $C=\frac{R}{\gamma-1}+\frac{R}{1-x}$
As the process is $\frac{P}{V}=$ **constant,**
i.e., PV^{-1} **constant, therefore,** $x=-1$**.**
For an ideal monatomic gas, $\gamma=\frac{5}{3}$
$\therefore C=\frac{R}{\frac{5}{3}-1}+\frac{R}{1-(-1)}=\frac{3}{2}R+\frac{R}{2}=2R$
$\Delta Q=nC(\Delta T)=1(2R)\left(2\ \mathrm{T}_0-\mathrm{T}_0\right)=2RT_0$

Q.15 Answer:

Potential difference at capacitor and inductor have phase difference of $\frac{\pi}{2}$.

Solution:

In a series L.C.R circuit, phasor is shown in the figure,



Where voltage drop across capacitor is shown by, $V_C = iX_C$

Where voltage drop across inductor is shown by, $V_L = iX_L$

Here i, X_C, X_L & V_R(in figure) represents current, capacitive reactance, inductive reactance and supply voltage respectively.

We can see in the figure that, current lead for voltage drop in capacitor by 90° and lag for voltage drop across inductor by 90°.

So, potential drop across capacitor and inductor is 180°.

Q.16 Answer:

Statement -1 is true, Statement-2 is false

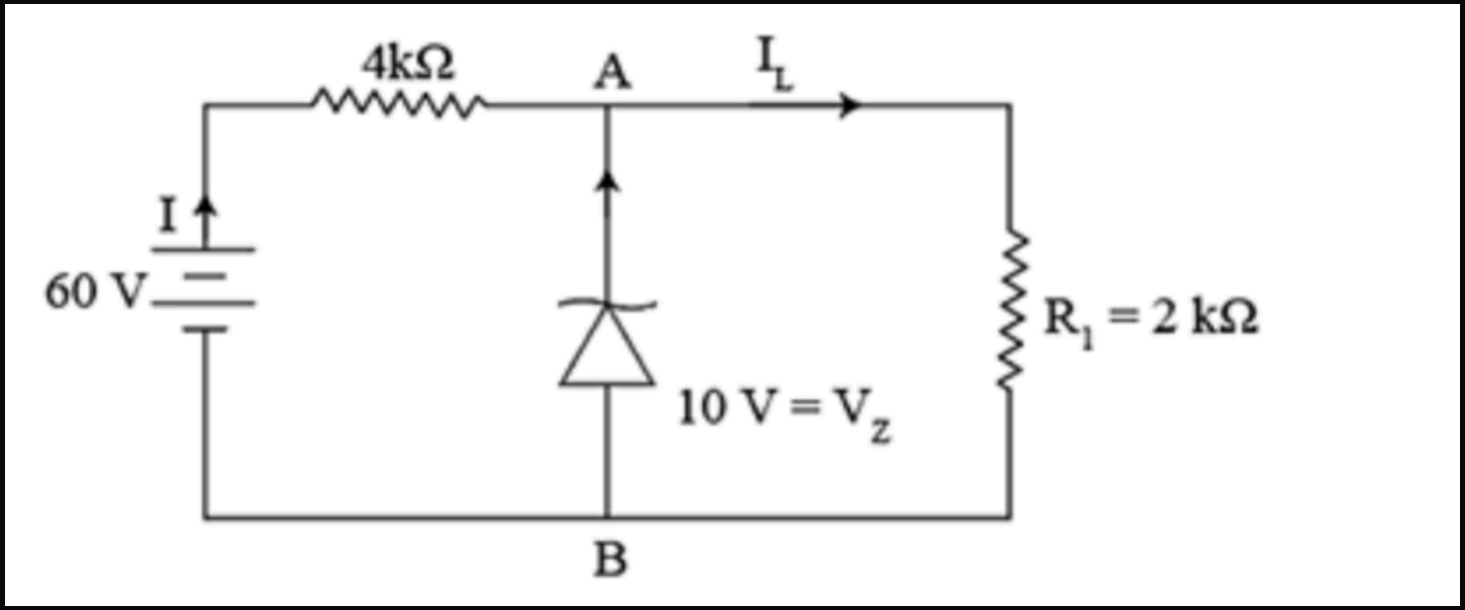
Solution:

Placing a soft iron core inside the coil makes the magnetic field radial which increase the torque acting on the coil due a give current flowing in the coil. The sensitivity increase because, for the same current, the deflection of the coil is more when a core is inserted inside the coil. Statement-1 is true and Statemenet-2 is false.

Q.17 Answer:

12.5 mA, 7.5 mA, 5 mA

Solution:



$$I_L = \frac{V_{AB}}{R_L} = \frac{10}{2000} = 5 \text{ mA}$$

$$I = \frac{60 - 10}{4000} = \frac{50}{4000} = 12 \cdot 5 \text{ mA}$$

$$I = I_L + I_Z \therefore I_Z = I - I_L = 7 \cdot 5 \text{ mA}$$

Hence, 12.5 mA, 7.5 mA, 5 mA

Q.18 Answer:

$\left(-\frac{R}{50}, \frac{21R}{80}\right)$

Solution:

Let, $M_2 = M, M_1 = 4M$

$$x_{cm} = \frac{4M(O) - M\left(\frac{R}{10}\right)}{4M + M}$$

$$\Rightarrow x_{cm} = -\frac{R}{50}$$

$$y_{cm} = \frac{4M\left(\frac{3R}{8}\right) - M\left(\frac{R}{16}\right)}{4M + M}$$

$$\Rightarrow y_{cm} = \frac{21R}{80}$$

Q.19 Answer:

0.1 m

Solution:

T = 0.06 N/m, $\rho = 1000 \text{ s/m}^3$ **; r = 0.1 cm**

msub $= 1100 \text{ N/m}^2; P = P_0 + 1100 \text{ N/m}^2$

We know $\Delta P = \frac{2T}{R}$

Now, $[P - (P_0 + \rho gh)] = \frac{2T}{R}$

$$\therefore \ (P - P_0) - \rho gh = \frac{2T}{R}$$

$$\therefore \ \rho gh = (P - P_0) - \frac{2T}{R}$$

$$= 1100 - \frac{2 \times 0.06}{0.1 \times 10^{-2}}$$

$$= 1100 - 120$$

$$h = \frac{980}{1000 \times 9.8} m$$

$$= \frac{980 \times 100}{1000 \times 9.8} cm$$

$$= 10 \text{ cm} = 0.1 \text{ m}$$

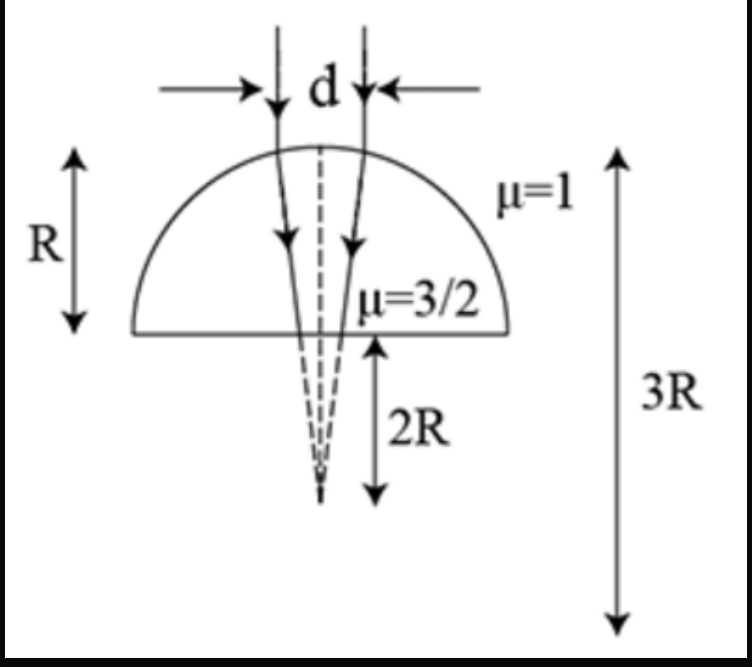
Q.20 Answer:

$\frac{2}{3}d$

Solution:

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \Rightarrow \frac{3}{2v} - \frac{1}{\infty} = \frac{\frac{3}{2} - 1}{R} \Rightarrow \frac{3}{2v} = \frac{1}{2R}$$

$$v = 3R \Rightarrow \frac{d}{3R} = \frac{d'}{2R} \Rightarrow d' = \frac{2}{3} d$$



Q.21 Answer:

—

Solution:

Phase difference $= \frac{\pi}{2}$**. Hence resultant amplitude**

$A = \sqrt{a_1^2 + a_2^2} = \sqrt{4^2 + 3^2} = 5$

Q.22 Answer:

—

Solution:

As given

$\left({}_1^2\mathrm{H}\right) + \left({}_1^2\mathrm{H}\right) \rightarrow \left({}_2^4\mathrm{H}\right) + \text{energy}$

The binding energy per nucleon of a deuteron $\left({}_1^2\mathrm{H}\right)$

= 1.1 MeV

$\Rightarrow$ **Total binding energy of one deuteron nucleus** $= 2 \times 1.1 = 2.2\ \mathrm{MeV}$

The binding energy per nucleon of helium = 7MeV

$\Rightarrow$ **Total binding energy** $4 \times 7 = 28\ \mathrm{MeV}$

$\Rightarrow$ **Hence Energy Released in the above process** $= 28 - 2 \times 2.2 = 23.6\ \mathrm{MeV}$

Q.23 Answer:

—

Solution:

n **= number of turns per unit length of solenoid ,**

$n = \frac{500}{1}$

$H = ni \Rightarrow i = \frac{H}{n} = \frac{4 \times 10^3}{500} = 8\ \mathrm{A}$

Q.24 Answer:

—

Solution:

Both upper half and lower half will have same effective area of $\frac{\pi R^2}{2}$ so charge in flux will be same and induced emf will have some value. But since the resistance is different due to which current must be different but ring is as a whole is closed circuit so electric field will be generated to make the current flow in both parts to be same.

$$\varepsilon + E\pi R - i10r = 0$$

$$\varepsilon - E\pi R - ir = 0 \quad \ldots(ii)$$

$$\Rightarrow \quad 2E\pi R - 9ir = 0$$

$$E = \frac{9\text{ir}}{2\pi R} \Rightarrow i = \frac{E}{11r} = \frac{\pi R^2 b}{11r}$$

$$\Rightarrow \quad E = \frac{9\text{r}}{2\pi R} \times \frac{\pi R^2 b}{11r} = \frac{9}{22}\text{Rb}$$

$$\therefore \quad \frac{9}{22}\text{Rb}$$

Q.25 Answer:

—

Solution:

$Y = \frac{\frac{F}{A}}{\frac{l}{L}} = \frac{F \times L}{A \times l}$ [l = Change in length, L = Original length] (where Y is Young's modulus of elasticity)

Since Y, L and A remain same.

$\Rightarrow F \propto l$

$\Rightarrow \frac{F_1}{F_2} = \frac{l_1}{l_2}$

$\Rightarrow \frac{F}{F_2} = \frac{2 \times 10^{-3}}{4 \times 10^{-3}}$

$\Rightarrow F_2 = 2F \Rightarrow F_2 = nF \Rightarrow n = 2$

Q.26 Answer:

$\text{NH}_2\,\text{CH}_2\,\text{COOH}$

Solution:

In aqueous solution, $\text{NH}_2\,\text{CH}_2\,\text{COOH}$, acts as a dipolar ion, hence it is a zwitter ion.

Amino acids is one of the best example of zwitter ions having a group of carboxyls (acidic in nature) and a group of amines (basic in nature).

Q.27 Answer:

Mg

Solution:

Lowest Standard Reduction Potential, highest reducing power. and metal will act as a reducing agent.

Q.28 Answer:

(a), (c)

Solution:

a) The angular momentum is given by $\frac{\text{nh}}{2\pi}$

whereas:

n corresponds to $K, L, M \ldots$ orbits.

High value of angular momentum means the value of n is large as $\frac{h}{2\pi}$ is a constant.

As the value of n increases, the distance of an electron from the nucleus also increases.

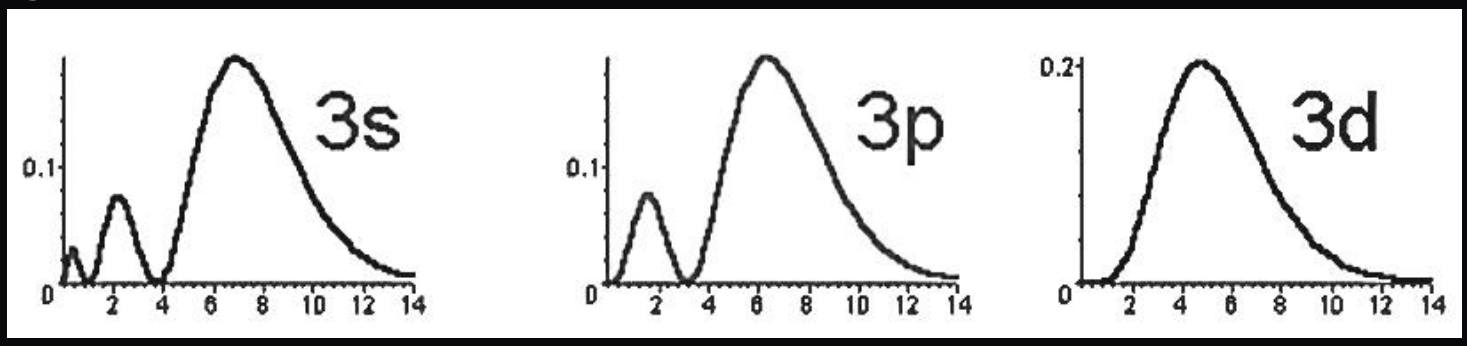
b) The principal quantum number depends on the size. As the value of n increases, the distance of an electron from the nucleus also increases but the azimuthal quantum number doesn't depend on size.

So, this is an incorrect statement.

c) The angular momentum is given by $\frac{nh}{2\pi}$.

For ground state, $n = 1$. So angular momentum=$\frac{h}{2\pi}$

d) The probability distribution curves give the probability of finding an electron in any direction from the nucleus. As the value of the Azimuthal quantum number increases, the peak shifts towards lower value, as shown in the figure.



Q.29 Answer:

Molecularity of a reaction is always a whole number

Solution:

Order of reaction is an experimental value can be whole number or fraction, while molecularity is a theoretical value and always natural number.

Q.30 Answer:

$D < A < C < B$

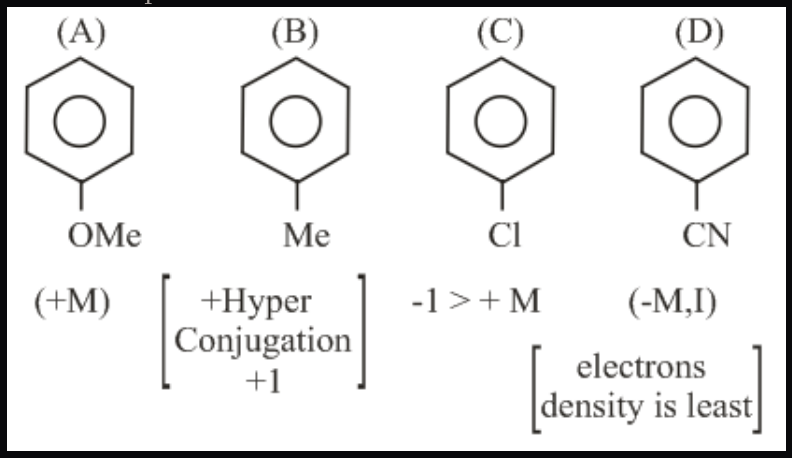
Solution:

Reactivity of E^+ substitution$\propto \pi e^-$ can be present on the aromatic compound.

Electrophilic substitution $\propto +M \propto \frac{1}{-M}$

$\propto +HC \propto \frac{1}{-HC}$

$\propto +I \propto \frac{1}{-I}$



(order of e^- density α electrophilic substitution)

Ring e^- density is highest.

So $B > C > A > D$

Q.31 Answer:

$P - IV,\ Q - I,\ R - II,\ S - III$

Solution:

Amino acids are monomers of proteins. These are substituted methanes and has a variable group R attached to the carbon. Due to presence of R group various types of amino acids are formed. Cysteine and Methionine are Sulphur containing amino acids. Tyrosine are amino acids with benzene ring. Glutamic acid is acidic and that means have acidic group. Serine has alcoholic group.

Q.32 Answer:

Blue to colourless

Solution:

Titration of iodine (I_2) by thiosulphate solution $\left(S_2O_3^{2-}\right)$ follows the reaction:

$2S_2O_3^{2-} + I_2 \rightarrow 2I^- + S_4O_6^{2-}$

In the presence of starch, free iodine present in the solution imparts blue colour to the solution.

End point marks the completion of reaction, that is, at this point all the iodine present in the solution is reduced to iodide ions by thiosulphate solution.

So, due to the absence of free iodine in the solution, the solution turns colourless.

Thus, the end point is indicated by colour change in the solution from blue to colourless.

Q.33 Answer:

Two unpaired electrons in antibonding MO

Solution:

Two unpaired electrons in antibonding MO

Q.34 Answer:

$-111.1\ \text{KJ mol}^{-1}$

Solution:

$(I) - (II) \Rightarrow$

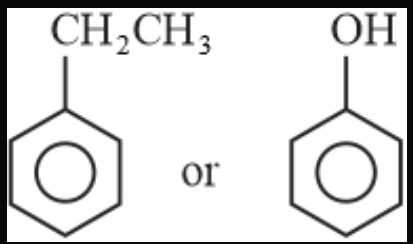
$C_{(s)} + \frac{1}{2}O_{2(g)} - CO_{(g)} = 0 \quad ; \quad \Delta H_{III} = -111.1\ \text{kJ}$

$C_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)} \quad ; \quad \Delta H_f^\circ = -111.1\ \text{kJ}$

Q.35 Answer:

2 and 4.

Solution:



will undergo Friedel Craft's alkylation on ortho or para position because of more electron density.

Q.36 Answer:

Statement-1, is True, Statement - 2 is True; Statement - 2 is a correct explanation for Statement-1.

Solution:

Vapour pressure is defined at the equilibrium state. Equilibrium state can be interpreted by the equilibrium constant. As Equilibrium constant change with temperature, vapour pressure will also change with temperature.

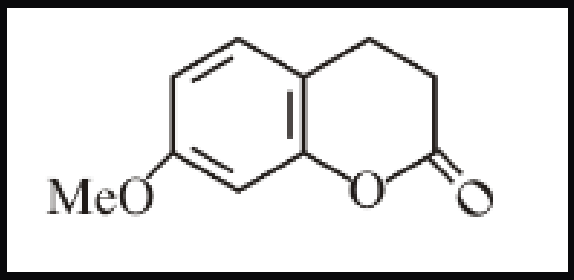
Q.37 Answer:

4 > 3 > 1 > 2

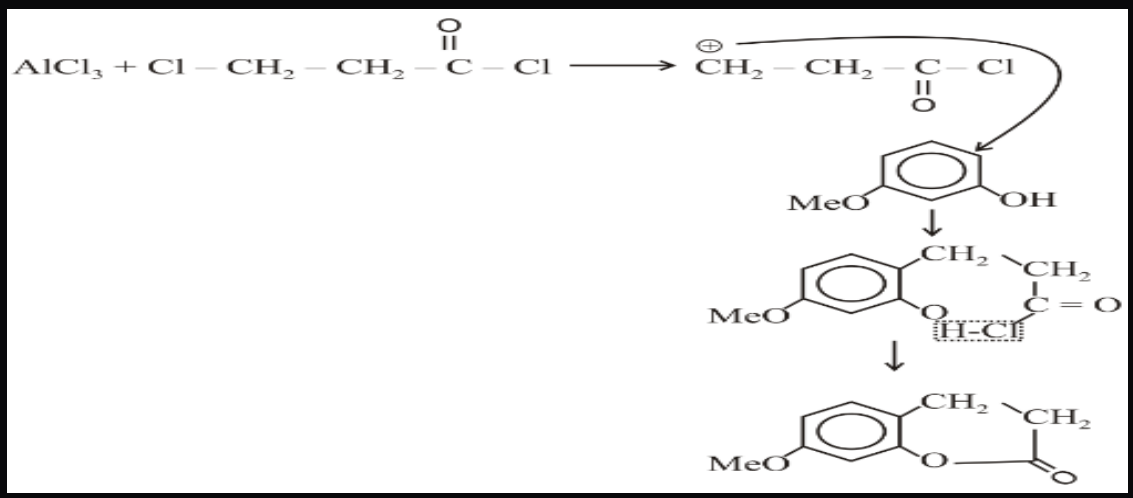
Solution:

$-CH_3$ group is inductively electron-donating and base-strengthening from all position. Moreover, if $-CH_3$ is attached to benzene ring (or if R has benzylic H), it is electron-donating by hyper-conjugation from the ortho and para positions. Because of ortho effect, o-substituted aniline become less basic than aniline. Thus,
$p-CH_3C_6H_4NH_2$ **(hyper-conjugation and induction) >** $m-CH_3C_6H_4NH_2$ **(induction) >**
$C_6H_5NH_2 > o-CH_3C_6H_4NH_2$ **(ortho effect).**

Q.38 Answer:



Solution:



Q.39 Answer:

Cerium (III) acts as an oxidising agent.

Solution:

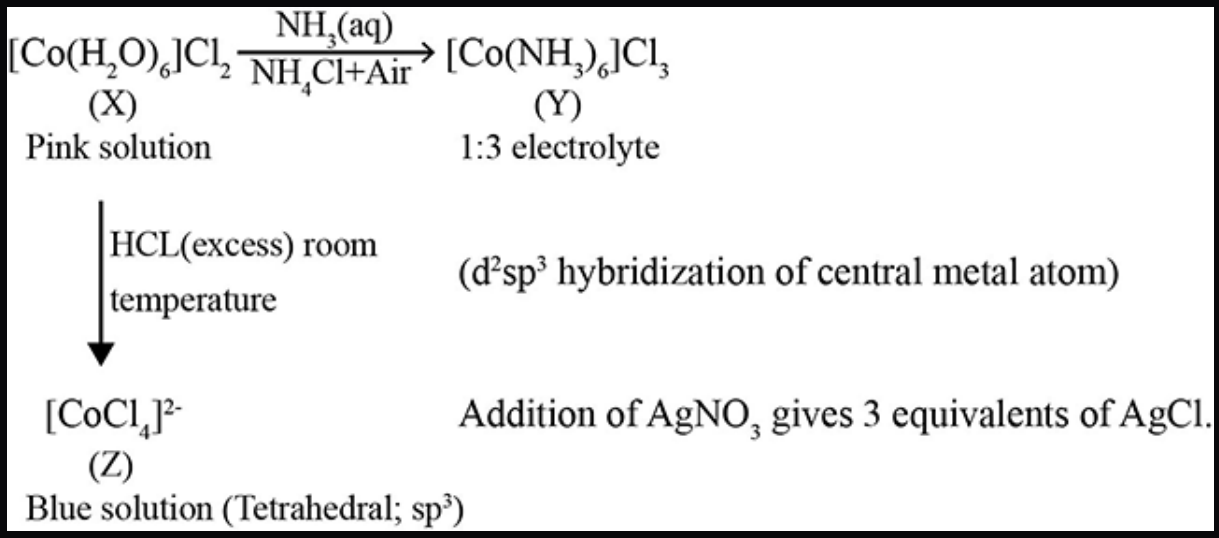
$+4$ oxidation state of cerium is also known in solutions.
$4f^1 5d^0 6S^2$
$4f^2 5d^0 6S^2$

The high charge on Ce^{4+} leads to high hydration, hydrated Ce^{4+} is hydrolysed giving polymeric species & H^+ . $Ce(4+)$ is stable in strongly acidic solutions & used as oxidising agent in volumetric analysis.

Q.40 Answer:

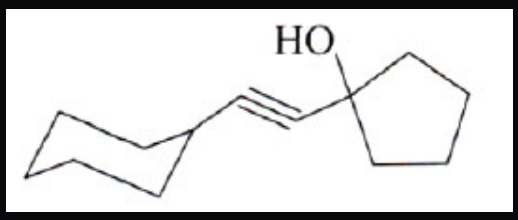
Addition of silver nitrate to Y gives only two equivalents of silver chloride

Solution:

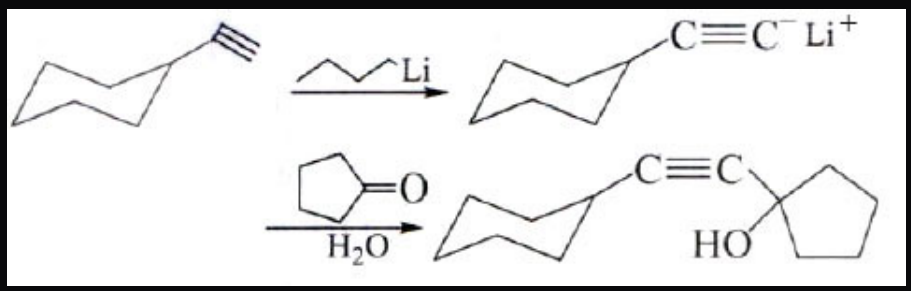


X and Z are in equilibrium at $0^\circ C$, equilibrium is shifted towards X, making colour of solution pink.

Q.41 Answer:



Solution:



Q.42 Answer:

11.3

Solution:

$pH = 2, [HCl] = 10^{-2}(M) \& pOH = 2$

or $[NaOH] = 10^{-2}(M)$

$4ml$ of $10^{-2}(M)HCl \equiv 4 \times 10^{-5}$ moles HCl.

$6ml$ of $10^{-2}(M)NaOH \equiv 6 \times 10^{-5}$ moles $NaOH$

After mixing total moles of $OH^- =$ moles of $NaOH -$ moles of $HCl = 6 \times 10^{-5} - 4 \times 10^{-5}$

After mixing excess moles of $OH^- = 2 \times 10^{-5}$ Total volume $= 6 + 4 = 10ml = 10 \times 10^{-3}$ litre

$[OH^-] = \frac{2\times10^{-5}}{10} \times 10^3 = 2 \times 10^{-3}$
or $\quad pOH = 3 - \log 2 = 3 - 0.3 = 2.7$
or $pH = 11.3$

Q.43 Answer:

If both Assertion and Reason are false statements, then mark (4).

Solution:

Hydrogen can be placed with Halogen gases as well it has one proton and one electron

Q.44 Answer:

Both Assertion and Reason are false statements.

Solution:

(i) In diamond, all C-atom are sp^3 hybridized.
(ii) In graphite hybridization is sp^2

Q.45 Answer:

i, ii & iii

Solution:

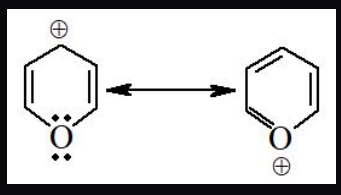
H.C. is the delocalisation of $\sigma -$ bond electrons.
Resonance is the delocalisation of $\pi -$ bond electrons.
Inductive effect is the partial displacement of $\sigma -$ bond electrons.
$\beta - H$ connected with Csp $(C \equiv C)$ So bond order increase hence bond energy increase (α) H remove the ion is stabilised by delocalisation while γ does not

Q.46 Answer:

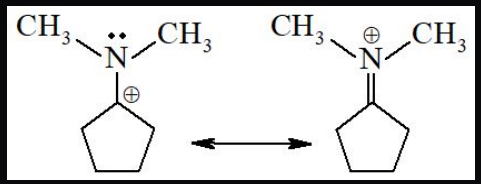
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Solution:

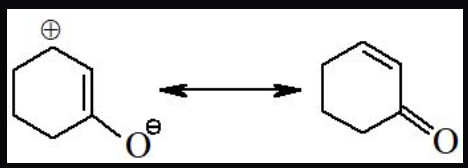
1.



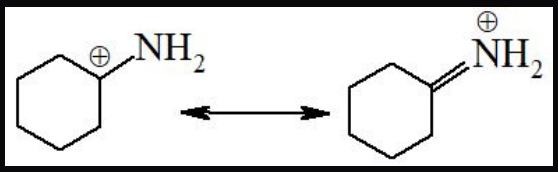
2.



3.



4.



Q.47 Answer:

—

Solution:

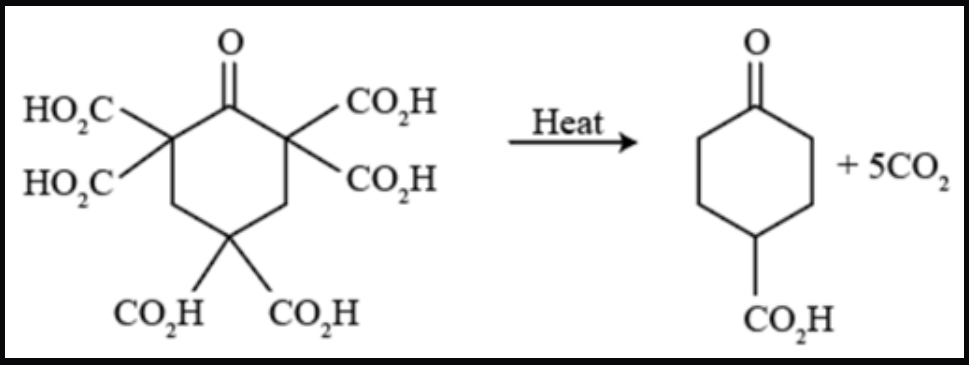
$[Co(en)_2Cl_2]^+ \longrightarrow$ will show cis-trans isomerism
$[CrCl_2\,(C_2O_4)_2]^{3-} \longrightarrow$ will show cis-trans isomerism
$[Fe\,(H_2O)_4\,(OH)_2]^+ \longrightarrow$ will show cis-trans isomerism
$[Fe(CN)_4\,(NH_3)_2]^- \longrightarrow$ will show cis-trans isomerism
$[Co(en)_2\,(NH_3)\,Cl]^{2+} \longrightarrow$ will show cis-trans isomerism
$[Co\,(NH_3)_4\,(H_2O)\,Cl]^{2+} \longrightarrow$ will not show cis-trans isomerism (although it will show geometrical isomerism)

Q.48 Answer:

—

Solution:

β-keto acids and Gem-dicarboxylic acids give CO_2 on heating.



Q.49 Answer:

—

Solution:

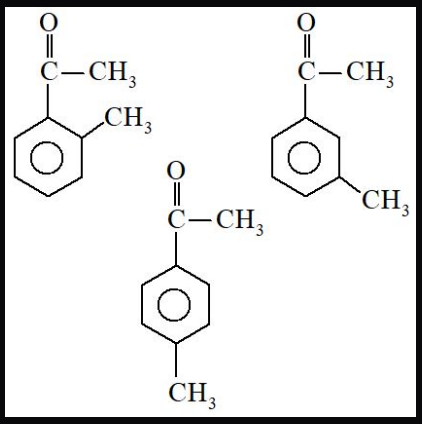
As for this reaction $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$
$\Delta S = \sum S_P - \sum S_R$
$\Delta S = 50 - \left(\frac{1}{2} \times 60 + \frac{3}{2} \times 40\right) = -40\ JK^{-1}mol^{-1}$
$\Delta G = \Delta H - T\Delta S$
Now as at equilibrium $\Delta G = 0$
So $\Delta H - T.\Delta S = 0$
Or $T = \frac{\Delta H}{\Delta S} = \frac{-30\times10^3}{-40}$
$T = 750\ K$

Q.50 Answer:

—

Solution:

Since A is reacting with the 2,4-DNP and showing iodoform reaction, it must be an aromatic methyl ketone.



Since all methyl ketones can give both 2,4-DNP test and iodoform test, Benzyl propanone will also give both tests.

Q.51 Answer:

$\frac{1}{4}$

Solution:

Given equation is

$\frac{dy}{dx} + y(-2\tan x) = \tan^2 x$

IF $= e^{2\ln(\cos x)} = \cos^2 x$

Thus, the solution is

$y \cdot \cos^2 x = \int \tan^2 x \cdot \cos^2 x dx$

$\Rightarrow y \cdot \cos^2 x = \int \frac{1-\cos 2x}{2} dx$

$= \frac{x}{2} - \frac{\sin 2x}{4} + C$

$\Rightarrow f(x) = \cos^2 x$

$\Rightarrow f\left(\frac{\pi}{3}\right) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

Q.52 Answer:

10

Solution:

Since, $\vec{a} \cdot \left(\vec{b} + \vec{c}\right) = 0$

$\Longrightarrow \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} = 0$

Similarly, $\vec{b} \cdot \vec{c} + \vec{b} \cdot \vec{a} = 0 \& \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} = 0$

$\therefore \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c} = 0 \quad \ldots(1)$

Given, $\left|\vec{a} + \vec{b}\right| = 6$

$\Longrightarrow \left|\vec{a}\right|^2 + \left|\vec{b}\right|^2 + 2\vec{a} \cdot \vec{b} = 36 \quad \ldots(2)$

Similarly, $\left|\vec{b}\right|^2 + \left|\vec{c}\right|^2 + 2\vec{b} \cdot \vec{c} = 64 \quad \ldots(2)$

$\& \left|\vec{c}\right|^2 + \left|\vec{a}\right|^2 + 2\vec{c} \cdot \vec{a} = 100 \quad \ldots(4)$

On adding Eqs. $(2), (3)$ **and** (4)**, we get**

$2\left|\overrightarrow{a}\right|^2+2\left|\overrightarrow{b}\right|^2+2\left|\overrightarrow{c}\right|^2+2\left(\overrightarrow{a}\cdot\overrightarrow{b}+\overrightarrow{b}\cdot\overrightarrow{c}+\overrightarrow{c}\cdot\overrightarrow{a}\right)=200$

$\Longrightarrow\left|\overrightarrow{a}\right|^2+\left|\overrightarrow{b}\right|^2+\left|\overrightarrow{c}\right|^2=100\quad\ldots(5)$ **[from Eqs.** (1)**]**

Now, $\left|\overrightarrow{a}+\overrightarrow{b}+\overrightarrow{c}\right|^2=\left|\overrightarrow{a}\right|^2+\left|\overrightarrow{b}\right|^2+\left|\overrightarrow{c}\right|^2+2\left(\overrightarrow{a}\cdot\overrightarrow{b}+\overrightarrow{b}\cdot\overrightarrow{c}+\overrightarrow{c}\cdot\overrightarrow{a}\right)$

$\Longrightarrow\left|\overrightarrow{a}+\overrightarrow{b}+\overrightarrow{c}\right|^2=100$ **[from Eqs.** (1) **and** (5)**]**

$\Longrightarrow\left|\overrightarrow{a}+\overrightarrow{b}+\overrightarrow{c}\right|=10$

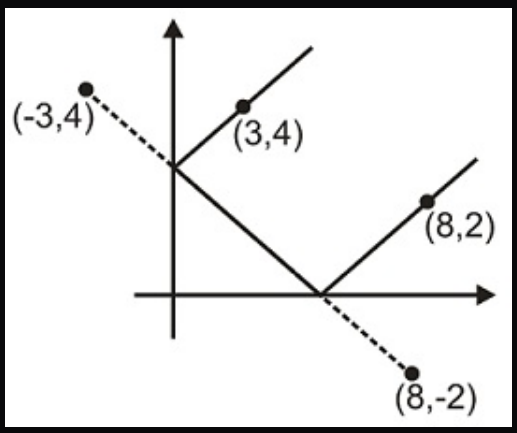
Q.53 Answer:

$x=4\frac{1}{3}$

Solution:

$\frac{0+2}{x-8}=\frac{4+2}{-3-8}=-\frac{6}{11}$

$x=\frac{13}{3}$



Q.54 Answer:

100

Solution:

$\because(1+x)^n=\sum_{r=0}^{n}{}^nC_rx^r=\sum_{r=0}^{n}a_rx^r$ **(given)**

$\therefore a_r={}^nC_r$

Also, $b_r=1+\frac{a_r}{a_{r-1}}=1+\frac{{}^nC_r}{{}^nC_{r-1}}=\frac{{}^{n+1}C_r}{{}^nC_{r-1}}$

So, $b_r=\left(\frac{n+1}{r}\right)$

$\because\ \prod_{r=1}^{n}b_r=\prod_{r=1}^{n}\left(\frac{n+1}{r}\right)=\frac{(n+1)^n}{n!}$

$=\frac{(101)^{100}}{100!}$ **(given)**

$\therefore n=100$

Q.55 Answer:

$\sqrt{(k_1-k_3)^2+(k_2-k_4)^2}$

Solution:

$L_1:x=k_1;\ y=k_2\ \&\ L_2:x=k_3;\ y=k_4$

Equation of these lines in symmetrical form is,

$L_1:\frac{x-k_1}{0}=\frac{y-k_2}{0}=\frac{z-0}{1}$

$L_2:\frac{x-k_3}{0}=\frac{y-k_4}{0}=\frac{z-0}{1}$

Since, both the lines have same direction ratios

$\therefore$ L_1 & L_2 **are parallel**

Shortest distance $d = \frac{\left|\overrightarrow{b} \times \left(\overrightarrow{a_2} - \overrightarrow{a_1}\right)\right|}{\left|\overrightarrow{b}\right|}$

$\overrightarrow{a_1} = k_1\hat{i} + k_2\hat{j} + 0\widehat{k}$

$\overrightarrow{a_2} = k_3\hat{i} + k_4\hat{j} + 0\widehat{k}$

$\overrightarrow{b} = 0\hat{i} + 0\hat{j} + \widehat{k}$

$$\overrightarrow{b} \times \left(\overrightarrow{a_2} - \overrightarrow{a_1}\right) = \begin{vmatrix} \hat{i} & \hat{j} & k \\ 0 & 0 & 1 \\ k_3 - k_1 & k_4 - k_2 & 0 \end{vmatrix}$$

$= (k_2 - k_4)\hat{i} - (k_1 - k_3)\hat{j}$

$\therefore\ d = \frac{\sqrt{(k_2 - k_4)^2 + (k_1 - k_3)^2}}{1}$

$d = \sqrt{(k_1 - k_3)^2 + (k_2 - k_4)^2}$

Q.56 Answer:

1

Solution:

Let A **and** D **be first term and common difference of the given AP respectively and** m **and** n **be the first term and common ratio of the given GP respectively.**

We have

$a = A + (p-1)D \ \ldots(i)$

$b = A + (q-1)D \ \ldots(ii)$

$c = A + (r-1)D \ \ldots(iii)$

and

$\mathrm{a} = \mathrm{mn}^{\mathrm{p}-1} \quad \ldots(\mathrm{iv}) \mathrm{b} \ = \mathrm{mn}^{\mathrm{q}-1} \ \ldots(\mathrm{v}) \mathrm{c} = \mathrm{mn}^{\mathrm{r}-1} \quad \ldots(\mathrm{vi})$

Product of the roots$= \frac{\mathrm{a}^{\mathrm{c}}\mathrm{b}^{\mathrm{a}}\mathrm{c}^{\mathrm{b}}}{\mathrm{a}^{\mathrm{b}}\mathrm{b}^{\mathrm{c}}\mathrm{c}^{\mathrm{a}}}$

$= \mathrm{a}^{\mathrm{c}-\mathrm{b}}\mathrm{b}^{\mathrm{a}-\mathrm{c}}\mathrm{c}^{\mathrm{b}-\mathrm{a}}$

Using $(i),\ (ii)$, **and**(iii)

$c - b = (r-q)D$

$a - c = (p-r)D$

$b - a = (q-p)D$

Hence the product of the roots

$= \left(\mathrm{mn}^{\mathrm{p}-1}\right)^{(\mathrm{r}-\mathrm{q})\mathrm{D}} \left(\mathrm{mn}^{\mathrm{q}-1}\right)^{(\mathrm{p}-\mathrm{r})\mathrm{D}} \left(\mathrm{mn}^{\mathrm{r}-1}\right)^{(\mathrm{q}-\mathrm{p})\mathrm{D}}$

$= \left[(\mathrm{m})^{(\mathrm{r}-\mathrm{q})\mathrm{D}+(\mathrm{p}-\mathrm{r})\mathrm{D}+(\mathrm{q}-\mathrm{p})\mathrm{D}}\right]\left[(\mathrm{n})^{(\mathrm{p}-1)(\mathrm{r}-\mathrm{q})\mathrm{D}+(\mathrm{q}-1)(\mathrm{p}-\mathrm{r})\mathrm{D}+(\mathrm{r}-1)(\mathrm{q}-\mathrm{p})\mathrm{D}}\right]$

$= m^0 n^0$

$= 1$

Q.57 Answer:

$-\frac{1}{3}$

Solution:

I.F.$= e^{-\int \frac{x}{1+x}dx} = e^{-\int \frac{x+1-1}{x+1}dx} = e^{-x+\ln|(x+1)|}$

$= \ e^{-x} \cdot |(x+1)|$

$y \cdot |x+1| e^{-x} = \int \frac{|x+1|}{x+1} e^{-x} dx + c$

$y \cdot \left|\left(x+1\right)\right| e^{-x} = -\frac{|\,x+1\,|}{x+1} e^{-x} + c$

$\Rightarrow\ y = -\frac{1}{x+1} + \frac{ce^{x}}{|\,x+1\,|}$

$x = 0,\ y = -1 \Rightarrow\ -1 = -1 + c \Rightarrow\ c = 0$

$\Rightarrow\quad y = -\frac{1}{1+x}$

$y(2) = -\frac{1}{3}$

Q.58 Answer:

$I_1 = I_2 < I_3$

Solution:

$f'\left(x\right) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = f'\left(0\right) + 2x$
$\Rightarrow f\left(x\right) = x^2 \Rightarrow I_1 = I_2 < I_3$

Q.59 Answer:

$5,\ 15$

Solution:

$f_2(x) = f(f(x)) = f(x) = x$
$f_3(x) = f(f_2(x)) = f(x) = x$
$\Rightarrow x^3 - 25x^2 + 175x - 375 = 0$
$\left(x-5\right)\left(x^2 - 20x + 75\right) = 0$
$(x-5)^2(x-15) = 0 \Rightarrow x = 5, 15$

Q.60 Answer:

$\frac{1}{6}$

Solution:

Let, $A = \frac{3}{2} < X < \frac{7}{2},\ B = X > 2$
$A \cap B = 2 < X < \frac{7}{2}$
$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{P(3)}{P(3)+P(4)+P(5)+P(6)}$
$= \frac{\frac{3}{12}}{\frac{3}{12}+\frac{4}{12}+\frac{5}{12}+\frac{6}{12}} = \frac{1}{6}$

Q.61 Answer:

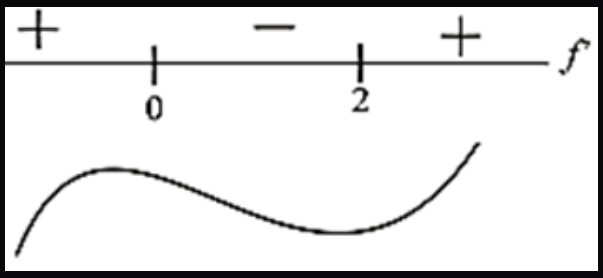
19

Solution:

$y = (x-1)^2 + b^3 - 3b^2 + 3$
$\therefore y_{\min} = f(b) = b^3 - 3b^2 + 3$

Now, $f'(b) = 3b^2 - 6b$

$= 3b(b-2)$



Also, $f(0) = 3$

$f(2) = -1$

$f(4) = 19$

$\therefore \max(f(b)) = 19$

Q.62 Answer:

2

Solution:

$\because \ (a^4 + b^4) = (a^2)^2 + (b^2)^2 = (a^2 + b^2)^2 - 2a^2b^2$

$\because \ (a^6 + b^6) = (a^2 + b^2)^3 - 3a^2b^2(a^2 + b^2)$

$\sin^6\alpha + \cos^6\alpha = (\sin^2\alpha + \cos^2\alpha)^3 - 3\sin^2\alpha\cos^2\alpha(\sin^2\alpha + \cos^2\alpha) = 1 - 3\sin^2\alpha\cos^2\alpha$

$\sin^4\alpha + \cos^4\alpha = (\sin^2\alpha + \cos^2\alpha)^2 - 2\sin^2\alpha\cos^2\alpha = 1 - 2\sin^2\alpha\cos^2\alpha$

$\sin^4\alpha + \cos^4\alpha = 1 - 2\sin^2\alpha\cos^2\alpha$

$\sin^6\alpha + \cos^6\alpha = 1 - 3\sin^2\alpha\cos^2\alpha$

$\therefore \ \frac{\sin^4\alpha+\cos^4\alpha+2\sin^2\alpha\cos^2\alpha}{\sin^6\alpha+\cos^6\alpha+3\sin^2\alpha\cos^2\alpha} + 1 = \frac{1-2\sin^2\alpha\cos^2\alpha+2\sin^2\alpha\cos^2\alpha}{1-3\sin^2\alpha\cos^2\alpha+3\sin^2\alpha\cos^2\alpha} + 1 = 2$

Q.63 Answer:

-1

Solution:

$\cos\alpha + i\sin\alpha$ **is a root of**

$a_n\left(\frac{1}{z}\right)^n + a_{n-1}\left(\frac{1}{z}\right)^{n-1} + \ldots + a_2\left(\frac{1}{z}\right)^2 + a_1\left(\frac{1}{z}\right) + 1 = 0$

.Equating real parts on both sides,

$a_n\cos n\alpha + a_{n-1}\cos(n-1)\alpha + \ldots + a_1\cos\alpha + 1 = 0$

Q.64 Answer:

$13\left(\frac{9}{7}\right)^{\frac{1}{6}}$

Solution:

Let $x_1, \ x_2 \ldots, \ x_6$ **are the observations and** $x_1 = 28$

$\Rightarrow \quad 28. \ x_2 \ldots x_6 = 13^6$

$\Rightarrow \quad x_2 \ldots x_6 = \frac{13^6}{28}$

Now correct observation is 36

$\Rightarrow \quad 36. \, x_2 \ldots x_6 = \frac{13^6}{28} \times 36$

So, now correct geometric mean $= 13\left(\frac{9}{7}\right)^{\frac{1}{6}}$

Q.65 Answer:

600

Solution:

$a + c = 2b$

So, both a and c will be even or both will be odd as their sum is even. In first 50 natural number 25 are even and 25 are odd. Required number of ways $^{25}C_2 + {}^{25}C_2 = 2 \times \frac{25 \times 24}{2} = 600$

Q.66 Answer:

400

Solution:

For observations $x_1,\ x_2,\ \ldots\ldots\ldots x_{50}$

Mean, $\bar{x} = \frac{\sum x_i}{50} = 16 \ldots\ldots$ **(i)**

Variance, $\sigma^2 = \frac{\sum x_i^2}{50} - \left(\bar{x}\right)^2 = 16^2$

$\Rightarrow \frac{\sum x_i^2}{50} = 16^2 + \left(\bar{x}\right)^2 = 16^2 + 16^2 = 512 \ldots\ldots$ **(ii)**

So, the mean value of $(x_1 - 4)^2,\ (x_2 - 4)^2,\ \ldots\ldots\ldots (x_{50} - 4)^2$ **will be**

$= \frac{\sum (x_i - 4)^2}{50} = \frac{\sum x_i^2 - 8\sum x_i + 16 \times 50}{50}$

$= \frac{\sum x_i^2}{50} - 8\frac{\sum x_i}{50} + 16 = 512 - 8 \times 16 + 16 = 400$ **(using (i) and (ii))**

Q.67 Answer:

$I + (2^n - 1)A$

Solution:

$\because\ \mathrm{A}^2 = \mathrm{A} \Rightarrow \mathrm{A}^3 = \mathrm{A}^4 = \mathrm{A}^n = \mathrm{A}$

Also, $(I + A)^n$

$$= I^n + {}^nC_1\, I^{n-1}\, A + {}^nC_2\, I^{n-2} A^2 + \ldots + {}^nC_n\, I^n\, A^n$$

$$= I + A[{}^nC_1 + {}^nC_2 + \ldots + {}^nC_n]$$

$$= I + A(2^n - 1)$$

Q.68 Answer:

$x^2 + y^2 = 1$

Solution:

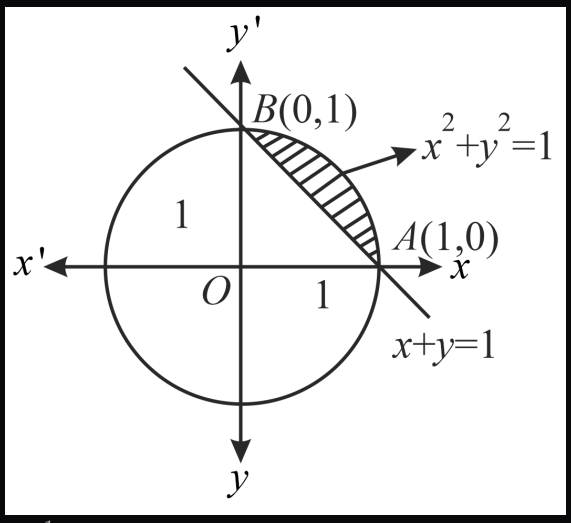
If e **and** e' **are the eccentricities of the hyperbola** $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ **and** $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ **, then the point** $\left(\frac{1}{e}, \frac{1}{e'}\right)$ **lies on the circle** $x^2 + y^2 = 1$ **because** $\frac{1}{e^2} + \frac{1}{(e')^2} = 1$**.**

Q.69 Answer:

$\left(\frac{\pi}{4} - \frac{1}{2}\right)$ sq unit

Solution:

Required area



$= \frac{1}{4}(\text{Area of circle}) - (\text{Area of } \Delta AOB)$

$= \frac{\pi(1)^2}{4} - \frac{1}{2} \times OA \times OB$

$= \left(\frac{\pi}{4} - \frac{1}{2}\right)$ **sq unit**

Q.70 Answer:

$\frac{x^2}{2} - \frac{4}{3}x^{\frac{3}{2}} + x + c$

Solution:

$\text{I} = \int \frac{(\sqrt{x}-1)^4}{(\sqrt{x}-1)^2} dx = \int (\sqrt{x}-1)^2 dx = \int (x - 2\sqrt{x} + 1) dx$

$= \frac{x^2}{2} - \frac{4}{3}x^{\frac{3}{2}} + x + c$

Q.71 Answer:

—

Solution:

Given, $^{100}C_{50} = \frac{100!}{50!\,50!} = \frac{100!}{(50!)^2}$

Now exponent of 2 **in** 100!

$= \left[\frac{100}{2}\right] + \left[\frac{100}{2^2}\right] + \left[\frac{100}{2^3}\right] + \left[\frac{100}{2^4}\right] + \left[\frac{100}{2^5}\right] + \left[\frac{100}{2^6}\right]$

$= [50] + [25] + [12.5] + [6.2] + [3.1] + [1.5]$

(Here $[*]$ **denotes** $G.I.F.$**)**

$= 50 + 25 + 12 + 6 + 3 + 1$

$= 97$

Now exponent of 3 **in** 100!

$= \left[\frac{100}{3}\right] + \left[\frac{100}{3^2}\right] + \left[\frac{100}{3^3}\right] + \left[\frac{100}{3^4}\right]$

$= [33.3] + [11.1] + [3.70] + [1.2]$

$= 33 + 11 + 3 + 1$; **Here** $[*]$ **denotes** $G.I.F.$

$= 48$

Thus exponent of 18 **in** $100! = 24$

Now exponent of 2 **in** 50!

$= \left[\frac{50}{2}\right] + \left[\frac{50}{2^2}\right] + \left[\frac{50}{2^3}\right] + \left[\frac{50}{2^4}\right] + \left[\frac{50}{2^5}\right]$

$= 47$

Now exponent of 18 **in** $50! = 11$

Hence, highest power of 18 **in** $^{100}C_{50}$

$= 24 - 22$

$= 2$

Q.72 Answer:

—

Solution:

We know that at any point on the XY-plane, the y-coordinate is zero.

Now, the vertices of the given triangle are $A(-1,1,1),\ B(1,-1,1)$ and $C(1,1,-1)$.

Hence, the coordinates of the vertices of the projected triangle will be $A'(-1,1,0),\ B'(1,-1,0)$ and $C'(1,1,0)$

Now, $\overrightarrow{A'B'} = (1+1)\hat{i} + (-1-1)\hat{j} + (0)\widehat{k} \Rightarrow \overrightarrow{A'B'} = 2\hat{i} - 2\hat{j}$ **and**

$\overrightarrow{A'C'} = (1+1)\hat{i} + (1-1)\hat{j} + (0)\widehat{k} \Rightarrow \overrightarrow{A'C'} = 2\hat{i}$

We know that, the area of the triangle $A'B'C' = \frac{1}{2}\left|\left(\overrightarrow{A'B'}\right) \times \left(\overrightarrow{A'C'}\right)\right|$

$= \frac{1}{2}\left|\left(2\hat{i} - 2\hat{j}\right) \times \left(2\hat{i}\right)\right| = \frac{1}{2}\left|4\widehat{k}\right|$

$= \frac{4}{2} = 2$ **square units.**

Q.73 Answer:

—

Solution:

Given that

$\underset{h\to\infty}{\text{Lim}} h^{-h^2}\left[(h+1)\left(h+\frac{1}{2}\right)\ldots\ldots\left(h+\frac{1}{2^{h-1}}\right)\right]^h$

$= \underset{h\to\infty}{\text{Lim}}\left[\frac{(h+1)\left(h+\frac{1}{2}\right)\ldots\ldots\left(h+\frac{1}{2^{h-1}}\right)}{h^h}\right]^h$

$= \underset{h\to\infty}{\text{Lim}}\left(\frac{h+1}{h}\right)^h\left(\frac{h+\frac{1}{2}}{h}\right)^h\ldots\ldots\left(\frac{h+\frac{1}{2^{h-1}}}{h}\right)^h$

$= \underset{h\to\infty}{\text{Lim}}\left(1+\frac{1}{h}\right)^h \cdot \left(1+\frac{1}{2h}\right)^{\frac{2h}{2}}\ldots\ldots\ldots\left(1+\frac{1}{2^{n-1}h}\right)^{\frac{2^{h-1}h}{2^{h-1}}}$

$= e^1 \cdot e^{\frac{1}{2}} \cdot e^{\frac{1}{4}}\ldots\ldots e^{\frac{1}{2^{n-1}}}\ldots\ldots\infty$

$= e^{1+\frac{1}{2}+\frac{1}{4}+\ldots\ldots\infty}$

$= e^{\frac{1}{1-\frac{1}{2}}} = e^2 = e^s \Rightarrow s = 2$

Q.74 Answer:

—

Solution:

$\mathrm{x}^2 - \mathrm{ky} + 32 = 0$

$\Rightarrow \mathrm{x}^2 = \mathrm{k}\left(\mathrm{y} - \frac{32}{\mathrm{k}}\right)$

Put, $\mathrm{x} = \mathrm{X}, \mathrm{y} - \frac{32}{\mathrm{k}} = \mathrm{Y}$

The equation of directrix is $\mathrm{Y} + \frac{\mathrm{k}}{4} = 0$

i.e. $\mathrm{y} - \frac{32}{\mathrm{k}} + \frac{\mathrm{k}}{4} = 0$

But, $\mathrm{y} - 2 = 0$ **is the directrix.**

$\Rightarrow \frac{32}{\mathrm{k}} - \frac{\mathrm{k}}{4} = 2$

$\Rightarrow \mathrm{k}^2 + 8\mathrm{k} - 128 = 0$

$\Rightarrow \mathrm{k} = -16$ **or** $\mathrm{k} = 8$

For $\mathrm{k} = 8$**, the parabola is** $\mathrm{x}^2 = 8(\mathrm{y} - 4)$ **which does not intersect the circle.**

For $k = -16$, the parabola is $x^2 = -16(y + 2)$ which intersects the circle at two real distinct points.

$\Rightarrow$ Absolute value of $k = |-16| = 16$

Q.75 Answer:

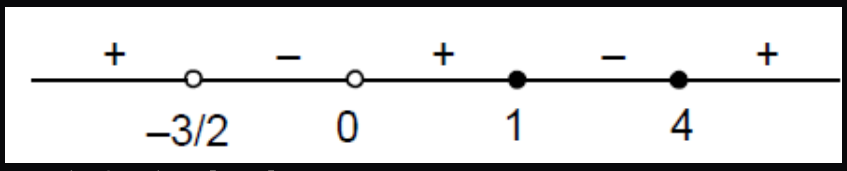
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Solution:

$\frac{(e^x-1)(2-\sin x)(x^2-5x+4)}{x^2(x^2-x+2)(2x+3)} \leq 0$

We can observe that $2 - \sin x > 0$ and $x^2 - x + 2 > 0 \, \forall \, x \in R$

so, the wavy curve is



$x \in \left(-\frac{3}{2}, \, 0\right) \cup \left[1, \, 4\right]$

So, the integral values of x are $-1, \, 1, \, 2, \, 3, \, 4$

Hence, $\lambda = 5$.